New Attacks on QR-UOV Across Vinegar Dimensions

Technical Supplement

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Appendix overview

The main body presents the direct forgery, directional equivalent-key recovery, characteristic-127 solver, and combined parameter consequences. The first appendices retain an independent smooth-projective route and two solver cross-checks. Later appendices give the complete characteristic-127 and concrete-cost derivations, deterministic equivalent-key fallbacks, direct-forgery proofs, expanded-key certificate calculations, seeded expansion, optional identification with the planted oil space, alternative low-space directions, and executable pseudocode. These details are separated so that readers can understand the attack and its parameter implications before entering the full audit.

A Independent projective smooth-core route

This section supplies the geometric input to Theorem B.18. Outside three explicit bad key loci, we show that some slice and output recombination gives a smooth 2^V-point projective core; we then bound the density of those loci. The next section chooses a finite working field and turns this existence statement into a verified randomized algorithm. Secret central coordinates are used only to expose the key distribution.

Let $K = \mathbb{F}_{127^3}$, $Q = |K|$, and

$$(V, O, m) \in \{(52, 18, 54), (76, 26, 78), (102, 35, 105)\}.$$

Put $e = m - V \in \{2, 2, 3\}$ and $N = V + O$. Work first in secret central coordinates

$$\mathcal{V} = K^V \oplus K^O, \qquad \mathcal{O}_0 = \{0\} \oplus K^O.$$

The m central quadrics are

$$F_i(v, o) = v^\mathsf{T} A_i v + 2v^\mathsf{T} B_i o, \qquad A_i \in \mathrm{Sym}_V(K),\ B_i \in K^{V\times O}. \tag{1}$$

In the public-random expansion model, the pairs (A_i, B_i) are independent and uniform. Equivalently, each F_i is uniform in

$$\mathcal{U} := H^0\Big(\mathbb{P}(\mathcal{V}), \mathcal{I}_{\mathbb{P}(\mathcal{O}_0)}(2)\Big), \qquad D_{\text{form}} := \dim \mathcal{U} = \frac{V(V+1)}{2} + VO.$$

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The proof uses central coordinates only to expose the distribution. The secret change from public to central coordinates lies in $\mathrm{GL}_N(K)$. Left multiplication by its inverse is a bijection on $L^{N\times(V+1)}$, so a uniform public slice remains uniform in central coordinates.

Smoothness, intersection multiplicity, and derivative-kernel recovery are coordinate invariant. In particular, the public and central tuples have identically vanishing joint discriminant polynomials simultaneously. Let

$$X_F := V(F_1, \dots, F_m) \subset \mathbb{P}(\mathcal{V}).$$

It contains the planted oil space $\mathbb{P}(\mathcal{O}_0) \simeq \mathbb{P}^{O-1}$.

Proposition A.1 (Exact marginal law on a fixed transverse slice)**.** Fix, independently of the sampled central forms, a K-rational projective V-plane Λ that meets $\mathbb{P}(\mathcal{O}_0)$ transversely in one point p. Under the marginal public-random law of the central pairs (A_i, B_i), the restrictions $F_1|_\Lambda, \dots, F_m|_\Lambda$ are independent uniform elements of

$$H^0\Big(\mathbb{P}^V, \mathcal{I}_p(2)\Big),$$

the K-vector space of projective quadrics through p.

Proof. Choose homogeneous coordinates $[z : t_1 : \cdots : t_V]$ on Λ with $p = [1 : 0 : \cdots : 0]$. Transversality makes the vinegar projection of Λ/p invertible, so after changing the t-coordinates its central-coordinate parametrization is

$$v = t, \qquad o = cz + Lt,$$

for some nonzero $c \in K^O$ and $L \in K^{O\times V}$. Hence

$$F_i|_\Lambda(z,t) = 2z\,t^\mathsf{T} B_i c + t^\mathsf{T}\Big(A_i + B_i L + L^\mathsf{T} B_i^\mathsf{T}\Big)\,t.$$

For every prescribed linear coefficient $\ell \in K^V$ and symmetric quadratic coefficient $H \in \mathrm{Sym}_V(K)$, the matrices B_i satisfying $B_i c = \ell$ form an affine space of the same size, and then $A_i = H - B_i L - L^\mathsf{T} B_i^\mathsf{T}$ is forced. Thus $(A_i, B_i) \mapsto F_i|_\Lambda$ is a surjective linear map with equal-size fibers. The pairs (A_i, B_i) are marginally independent and uniform by Proposition 3.3, proving the claim. □

Remark A.2 (Why the statement is marginal). The model permits the secret graph G to depend arbitrarily on A. Conditioning on G can therefore change the law of A, so no conditional-on-G uniformity is asserted. None is needed: the bad-key events in this section depend only on the marginal central tuple $(A_i, B_i)_i$, while a fresh public slice remains uniform after the fixed secret coordinate change for every individual key.

Proposition A.3 (Generic isolation by the spare equations)**.** Fix $p \in \mathbb{P}^V$ and let $m > V$. A Zariski-generic m-tuple in $H^0(\mathbb{P}^V, \mathcal{I}_p(2))^m$ has p as its only common projective zero. Generically its full Jacobian has rank V at p.

Proof. For each $x \neq p$, evaluation at x is a nonzero linear functional on $H^0(\mathbb{P}^V, \mathcal{I}_p(2))$. Requiring all m forms to vanish at x therefore imposes codimension m. The incidence with $x \in \mathbb{P}^V \setminus \{p\}$ has dimension at most the parameter-space dimension plus $V - m$, strictly smaller than the parameter-space dimension. By Chevalley's theorem, its image is constructible and its Zariski closure has dimension no larger than the incidence, so that closure is proper. Hence a generic tuple has no common zero away from p. At p, the first derivatives of a uniform quadric through p range over the full cotangent space. The rank-deficient $m \times V$ derivative matrices form a proper determinantal subvariety. □

The two propositions explain why the overdetermined sliced system generically contains only the planted point. The main attack nevertheless solves a square core because Proposition A.3 is qualitative: it does not supply the finite-field density and solver interface needed for a quantified faster theorem.

Let L/K be the finite field chosen below. One invocation samples

$$M \leftarrow L^{N\times(V+1)}.$$

If $\operatorname{rank} M < V+1$, the trial fails. Otherwise the attack restricts the public quadrics to the projective V-plane $\Lambda_M = \mathbb{P}(\operatorname{im} M)$:

$$q_i(t) = F_i(Mt), \qquad t \in \mathbb{P}^V_L.$$

It next samples $R \leftarrow L^{V\times m}$. A rank defect again ends the trial; for full-rank R, the square core is

$$g_a(t) = \sum_{i=1}^{m} R_{a,i} q_i(t), \qquad 1 \le a \le V.$$

Rank defects are counted as failed draws rather than conditioned away. This keeps the entries of (M, R) uniform, so the Schwartz–Zippel bound applies directly. By Lemma A.5, every rank defect already lies in $\{\Theta_F = 0\}$.

In secret affine coordinates, a transverse Λ_M is exactly a slice $o = c + Lv$ after translating its unique oil intersection. The projective parameterization is preferable in the proof because it has no inverses or denominators: every restricted coefficient is simply quadratic in the entries of M.

A.1 Why the full discriminant is enough

The finite-field Kronecker theorem used below assumes that every affine prefix is a reduced complete intersection. A smooth full projective core already implies these prefix conditions; no product of prefix discriminants is needed.

Lemma A.4 (A smooth full core gives reduced regular prefixes)**.** Let $g_1, \ldots, g_V$ be homogeneous quadrics in $\mathbb{P}^V$ over a field of odd characteristic. If their common zero scheme is a smooth zero-dimensional complete intersection, then the ordered quadrics form a regular sequence and every homogeneous prefix ideal $(g_1, \ldots, g_s)$ is geometrically radical. After any projective coordinate change and dehomogenization in an affine chart containing a final root, the affine sequence remains regular and every prefix ideal remains geometrically radical. These are precisely the solver's hypotheses with localization polynomial equal to 1.

Proof. Work over an algebraic closure and let S be the homogeneous coordinate ring. The final ideal has height V and is generated by V elements. Since S is Cohen–Macaulay, $g_1, \ldots, g_V$ is a regular sequence. Every prefix $I_s = (g_1, \ldots, g_s)$ is therefore an unmixed complete intersection, and its coordinate ring is Cohen–Macaulay.

Suppose that some prefix were not radical. A Cohen–Macaulay ring satisfies Serre's condition (S_1), so a nonreduced prefix cannot be generically reduced: some irreducible component occurs in its complete-intersection cycle with multiplicity at least two. Each remaining quadric is a nonzerodivisor modulo the preceding prefix and hence contains no component. Until the final cut, every surviving component has positive projective dimension, so every remaining positive-degree hypersurface meets it. The associativity formula for intersection multiplicities preserves the component's multiplicity factor through these proper cuts. The final zero-cycle would therefore contain a point with multiplicity at least two, contradicting smoothness of the final scheme. Thus every homogeneous prefix is geometrically radical.

Localization at a chart equation and dehomogenization preserve regular sequences and reducedness. The same conclusions hold in every affine chart containing a final root; roots at infinity are simply absent from the affine zero scheme. $\square$

Let $\Delta_{V,V}$ be the reduced discriminant for V quadrics in $\mathbb{P}^V$. It vanishes exactly when the common zero scheme is not a smooth complete intersection of codimension V. Because $\operatorname{char} K = 127 \neq 2$, Benoist's arbitrary-characteristic formula [6] gives the partial degree

$$d_V = (V+1)2^{V-1} \tag{2}$$

in the coefficients of each equation. Thus it suffices to make the single polynomial

$$\Theta_F(M,R) := \Delta_{V,V}\left(\sum_i R_{1i}F_i(Mt), \ldots, \sum_i R_{Vi}F_i(Mt)\right) \tag{3}$$

nonzero. Each coefficient of a restricted equation has degree at most two in M and one in the corresponding row of R. Since the discriminant has degree d_V in the coefficients of each of the V equations,

$$\deg_{M,R} \Theta_F \leq (2V+V)d_V = 3V(V+1)2^{V-1}. \tag{4}$$

Lemma A.5 (Nonvanishing certifies full parameter rank)**.** If $\Theta_F(M,R) \neq 0$, then $\operatorname{rank} M = V+1$ and $\operatorname{rank} R = V$.

Proof. If $\operatorname{rank} M < V+1$, then $\mathbb{P}(\ker M)$ is nonempty and every pull-back $F_i(Mt)$, together with all of its first derivatives, vanishes there; the core is singular. If $\operatorname{rank} R < V$, the output equations are linearly dependent and cannot cut a smooth codimension-V complete intersection. In either case the full discriminant vanishes. $\square$

A.2 Geometric nonvanishing of the joint slice-and-output discriminant

For $x \in X_F$, let

$$\rho_F(x) := \operatorname{rank}\big(dF_1(x), \ldots, dF_m(x)\big),$$

where the differentials are taken on $T_x\mathbb{P}(\mathcal{V})$.

Definition A.6 (Three key-level bad loci)**.** Define:

$$\begin{aligned}
\mathcal{B}_{\text{oil}} &:= \{\, F : \rho_F(p) < V \text{ for every } p \in \mathbb{P}(\mathcal{O}_0)(\overline{K})\},\\
\mathcal{B}_{\text{off}} &:= \{\, F : \exists x \in X_F(\overline{K}) \setminus \mathbb{P}(\mathcal{O}_0) \text{ with } \rho_F(x) < V\},\\
\mathcal{B}_{\text{span}} &:= \{\, F : \dim \operatorname{span}(F_1, \ldots, F_m) \leq V\}.
\end{aligned}$$

The next three lemmas separate the proof into linear-section geometry, a second-order separability argument, and generic output mixing.

Lemma A.7 (A transverse finite base slice through a regular oil point)**.** Assume $F \notin \mathcal{B}_{\text{oil}} \cup \mathcal{B}_{\text{off}}$. Choose $p \in \mathbb{P}(\mathcal{O}_0)(\overline{K})$ with $\rho_F(p) = V$. Then a Zariski-open dense set of projective V-planes $\Lambda \subset \mathbb{P}(\mathcal{V}_{\overline{K}})$ through p has all of the following properties:

1. $\Lambda \cap \mathbb{P}(\mathcal{O}_0) = \{p\}$;
2. $X_F \cap \Lambda$ is finite;
3. at every $x \in X_F \cap \Lambda$, the restricted differential $T_x\Lambda \to \overline{K}^m$ has rank V.

Proof. **Dimension of the base locus.** First, $\dim X_F \leq O - 1$. Otherwise choose a closed off-oil point x on a component of dimension at least O. The Zariski tangent space is the common differential kernel and has dimension at least the local dimension, so

$$\rho_F(x) = (N - 1) - \dim T_x X_F \leq V - 1,$$

contrary to $F \notin \mathcal{B}_{\text{off}}$. Since $\mathbb{P}(\mathcal{O}_0) \subset X_F$, the maximal dimension is exactly $O - 1$.

Behavior at the planted point. At p, every differential annihilates $T_p\mathbb{P}(\mathcal{O}_0)$. Both this tangent space and $\ker dF(p)$ have dimension $O-1$, so they are equal. Therefore a general V-plane through p is transverse at p and intersects the oil space only in p. We henceforth restrict to this nonempty open subset of the Grassmannian; no other oil point can occur in the slice.

Off-oil points not lying on a base line. Let $\mathcal{G}_p$ be the Grassmannian of projective V-planes through p, of dimension $V(O-1)$. We now consider only $x \in X_F \setminus \mathbb{P}(\mathcal{O}_0)$. Put $K_x = \ker dF(x)$; the off-oil hypothesis gives $\dim K_x \leq O - 1$. For fixed x, planes through p and x form a family of dimension $(V-1)(O-1)$. If the tangent direction of the line $\overline{px}$ is not in K_x, the additional condition $T_x\Lambda \cap K_x \neq 0$ is a proper Schubert condition and lowers this dimension by at least one. Allowing x to vary in dimension at most $O - 1$ gives total dimension at most $V(O-1) - 1$.

Lines through the planted point. If the tangent direction of $\overline{px}$ lies in K_x, then every quadric F_i vanishes identically on that line: both endpoint values and the polar cross term vanish. Each such direction contributes a full projective line contained in X_F, so passing from points of X_F to line directions lowers dimension by at least one. The direction variety therefore has dimension at most $\dim X_F - 1 \leq O - 2$. Planes containing one such line again form a family of dimension at most

$$(O-2) + (V-1)(O-1) = V(O-1) - 1.$$

Conclusion. The off-oil rank-defect incidence is locally closed and has dimension at most $V(O-1) - 1$ by the two cases above. Its closure in the projective product $X_F \times \mathcal{G}_p$ has the same dimension, and the projection of that closure is therefore a proper closed subset of $\mathcal{G}_p$. Outside it, every intersection point has restricted derivative rank V. At any such point the Zariski tangent space of $X_F \cap \Lambda$ is $T_x\Lambda \cap \ker dF(x) = 0$. Since local dimension never exceeds tangent-space dimension, no point can lie on a positive-dimensional component. Hence the intersection is finite. □

Lemma A.8 (A second-order spare equation forces separability)**.** Let $f_1, \dots, f_m$ be quadrics on $\mathbb{P}^V$ with a common point p. Suppose their differential matrix at p has rank V. If there is a nonzero

$$h \in \operatorname{span}(f_1, \dots, f_m)$$

whose first jet at p is zero, then the rational map

$$\phi_f : \mathbb{P}^V \dashrightarrow \mathbb{P}^{m-1}, \qquad x \longmapsto [f_1(x) : \cdots : f_m(x)]$$

has differential rank V on a nonempty open subset.

Proof. Choose projective coordinates with $p = [1 : 0 : \cdots : 0]$ and affine coordinates $y = (y_1, \dots, y_V)$. After an equation-basis change, the first V forms have expansions

$$f_j(y) = y_j + q_j(y),$$

where each q_j is quadratic. Since h is a quadric with zero value and zero differential at p, it is a nonzero homogeneous quadratic $q(y)$ on this chart.

Consider the $(V+1) \times (V+1)$ first-jet minor whose rows are $f_1, \dots, f_V, h$ and whose columns are value followed by the V first derivatives. Up to a nonzero sign, its lowest-degree homogeneous term is

$$q(y) - \nabla q(y) \cdot y = -q(y),$$

using Euler's identity $\nabla q(y) \cdot y = 2q(y)$. It is therefore nonzero. Hence the full first-jet rank is $V + 1$ generically, which is equivalent to differential rank V for ϕ_f away from the base locus. $\square$

Lemma A.9 (Finite regular base locus plus separability)**.** Let $f_1, \dots, f_m$ be quadrics on $\mathbb{P}^V$ over an algebraically closed field. Assume:

1. their common base scheme $B = V(f_1, \dots, f_m)$ is finite and nonempty;
2. the $m \times V$ projective differential matrix has rank V at every point of B;
3. ϕ_f has differential rank V on a nonempty open subset.

Then a nonempty Zariski-open set of matrices $R \in k^{V \times m}$ makes

$$\left(\sum_i R_{1i} f_i, \dots, \sum_i R_{Vi} f_i \right)$$

a smooth zero-dimensional complete intersection of degree 2^V.

Proof. Let $\mathcal{R} = k^{V \times m}$. For $x \notin B$, write

$$a(x) = (f_1(x), \dots, f_m(x)) \neq 0$$

and let $J(x)$ be the differential matrix. A matrix R has x as a root exactly when $Ra(x) = 0$, a codimension-V linear condition on $\mathcal{R}$.

On the open set where the first-jet matrix $[a(x) \mid J(x)]$ has rank $V + 1$, the map

$$a(x)^\perp \longrightarrow T_x^* \mathbb{P}^V, \qquad r \longmapsto rJ(x)$$

is surjective. Consequently the additional condition $\det(RJ(x)) = 0$ has codimension one inside $Ra(x) = 0$. After allowing x to vary in a V-dimensional space, this singular-root incidence still has codimension at least one in $\mathcal{R}$.

The first-jet rank-defect locus away from B has dimension at most $V-1$ by assumption 3. There the root condition $Ra(x)=0$ alone has codimension V, so its total incidence also has dimension at most $\dim\mathcal{R}-1$.

Finally, at each $x \in B$, the root condition is automatic, but the rank-V derivative assumption makes $\det(RJ(x))=0$ a genuine hypersurface in $\mathcal{R}$. Since B is finite, these basepoint hypersurfaces do not fill $\mathcal{R}$.

Globally, the singular-root incidence is the closed subset of $\mathbb{P}^V \times \mathcal{R}$ cut out by $Ra(x)=0$ and the maximal minors of $RJ(x)$. Its projection is closed because $\mathbb{P}^V$ is proper, and the preceding stratified dimension bounds show that this projection is proper. Choose R outside it. The resulting projective zero scheme is nonempty because it contains B. It cannot have positive dimension: at every root the Jacobian has rank V, so the Zariski tangent space is zero-dimensional. Therefore the zero scheme is smooth and zero-dimensional. Since it is cut out by V quadrics in $\mathbb{P}^V$, it is a complete intersection of degree 2^V. □

Theorem A.10 (Existence of a smooth projective core)**.** If

$$F \notin \mathcal{B}_{\text{oil}} \cup \mathcal{B}_{\text{off}} \cup \mathcal{B}_{\text{span}},$$

then the joint polynomial $\Theta_F(M,R)$ in (3) is nonzero.

Proof. Choose a regular geometric oil point $p \in \mathbb{P}(\mathcal{O}_0)(\overline{K})$. By Lemma A.7, a dense open subset of the irreducible Grassmannian of V-planes through p has finite base intersection and full restricted derivative rank at every basepoint.

After scalar extension, let

$$T_{F,\overline{K}} : \overline{K}^m \longrightarrow \mathcal{U}_{\overline{K}}, \qquad (\lambda_i)_i \longmapsto \sum_i \lambda_i F_i.$$

The coefficient kernel of the derivative map at p has dimension $m-V=e$ over $\overline{K}$. Scalar extension preserves the rank of the coefficient map, so

$$\dim_{\overline{K}} \ker T_{F,\overline{K}} = m - \dim_K \operatorname{span}_K(F_1,\dots,F_m).$$

This relation space lies inside the derivative kernel. Since $F \notin \mathcal{B}_{\text{span}}$, its dimension is strictly smaller than e. We may therefore choose a derivative-kernel vector outside $\ker T_{F,\overline{K}}$; it produces a nonzero global quadric $H \in \operatorname{span}_{\overline{K}}(F_i)$ with zero first jet at p.

The condition $H|_\Lambda \not\equiv 0$ is open and nonempty: if H vanished on every V-plane through p, it would vanish on their union, which is all of $\mathbb{P}(\mathcal{V})$. Intersecting this open set with the one supplied by Lemma A.7 gives a plane Λ satisfying both properties. On this plane, Lemma A.8 makes the restricted linear system generically separable, and Lemma A.9 supplies an output matrix R whose core is a smooth complete intersection of degree 2^V. Hence Θ_F is not the zero polynomial. □

A.3 Finite-field density of the three bad loci

A.3.1 First-jet surjectivity away from oil

Lemma A.11 (The central-form space spans every off-oil first jet)**.** For every $x \in \mathbb{P}(\mathcal{V}) \setminus \mathbb{P}(\mathcal{O}_0)$, the first-jet evaluation map

$$\mathcal{U} \longrightarrow K \oplus T_x^*\mathbb{P}(\mathcal{V})$$

is surjective after scalar extension to $\overline{K}$.

Proof. Choose a vinegar coordinate v_j that is nonzero at x and normalize to the chart $v_j = 1$. The quadrics

$$v_j^2, \quad v_j v_k \ (k \neq j), \quad v_j o_a \ (1 \leq a \leq O)$$

belong to $\mathcal{U}$ and restrict to the affine functions 1, the remaining vinegar coordinates, and the oil coordinates. Their first jets form a basis of the value-and-cotangent space. $\square$

A.3.2 The off-oil incidence

The codimension count has a simple structure. At a fixed off-oil point, the values and first derivatives of each central form can be prescribed independently by Lemma A.11. Making the point a common zero costs m conditions, and forcing derivative rank at most $V-1$ adds the usual determinantal codimension. Projecting away the point can recover at most its $N-1$ coordinates. The next lemma makes this count uniform and controls the projection degree.

Let $\mathcal{A} = \mathcal{U}^m$ be the affine key space. For an affine variety, write cdeg for the sum of the degrees of its irreducible components. For each vinegar chart $C_j \simeq \mathbb{A}^{N-1}$, let $Z_j \subset \mathcal{A} \times C_j$ be the incidence of pairs (F, x) such that all $F_i(x)$ vanish and the $m \times (N-1)$ derivative matrix has rank at most $V-1$.

Lemma A.12 (Codimension and cumulative degree of the off-oil incidence)**.** Each Z_j is pure of codimension

$$m + c_{\text{det}}, \qquad c_{\text{det}} = (m - V + 1)O = (e+1)O,$$

and satisfies

$$\operatorname{cdeg}(Z_j) \leq 3^m (2V)^{c_{\text{det}}}.$$

If Y_j is the Zariski closure of its projection to key space, then

$$\operatorname{codim}_{\mathcal{A}} Y_j \geq c_{\text{off}} := m + c_{\text{det}} - (N-1) = e(O+1) + 1, \qquad \operatorname{cdeg}(Y_j) \leq \operatorname{cdeg}(Z_j).$$

Proof. **Structure and codimension.** On the chart $v_j = 1$, the monomials v_j^2, $v_j v_k$ for $k \neq j$, and $v_j o_a$ restrict to 1, the remaining vinegar coordinates, and the oil coordinates. They give a polynomial splitting of value-and-first-jet evaluation. After a triangular polynomial change of coefficient coordinates, Z_j is the product of an affine space, the equations setting m value coordinates to zero, and the determinantal variety of $m \times (N-1)$ matrices of rank at most $V-1$. Hence Z_j is irreducible and pure of codimension

$$m + (m - V + 1)((N-1) - (V-1)) = m + (m - V + 1)O.$$

Cumulative degree. In the original coefficient-and-point coordinates, every value equation has total degree at most three. Every derivative entry has degree at most two, so each $V \times V$ minor has degree at most $2V$. At a generic rank-$(V-1)$ point, the determinantal variety is smooth of codimension c_{det}, and the differentials of its maximal minors span the conormal space. Choose c_{det} generic linear combinations of those minors whose differentials form a basis there. Together with the m value equations, they cut a complete intersection of codimension $m + c_{\text{det}}$ that contains the irreducible variety Z_j as a component. Cumulative Bézout therefore gives

$$\operatorname{cdeg}(Z_j) \leq 3^m (2V)^{c_{\text{det}}}$$

by the cumulative-degree bounds of Heintz and Lachaud–Rolland [24, 25].

Projection to key space. Projection can recover at most the $N-1$ chart coordinates, which gives

$$\operatorname{codim}_{\mathcal{A}} Y_j \geq m + c_{\text{det}} - (N-1) = e(O+1) + 1.$$

The coordinate projection $\mathcal{A} \times C_j \to \mathcal{A}$ is linear. The standard degree-under-linear-projection inequality gives

$$\deg \overline{\pi(Z_j)} \leq \deg Z_j$$

for an irreducible affine variety [24, 25]. Since both Z_j and its image closure Y_j are irreducible, this is exactly $\mathrm{cdeg}(Y_j) \leq \mathrm{cdeg}(Z_j)$. □

Proposition A.13 (Explicit off-oil density)**.** Under the ideal-expansion distribution,

$$\Pr[F \in \mathcal{B}_{\text{off}}] \leq \eta_{\text{off}} := V\, 3^m (2V)^{(e+1)O} Q^{-\{e(O+1)+1\}}. \tag{5}$$

Proof. Every off-oil point lies in one of the V vinegar charts, so $\mathcal{B}_{\text{off}} \subseteq \bigcup_j Y_j$. The affine finite-field point bound

$$|Y_j(\mathbb{F}_Q)| \leq \mathrm{cdeg}(Y_j) Q^{\dim Y_j}$$

from Lachaud–Rolland [25], followed by the union bound and Lemma A.12, gives the formula after division by $|\mathcal{A}(\mathbb{F}_Q)|$. □

A.3.3 The oil and low-span loci

Let

$$\tau_{m,V} = 1 - \prod_{j=0}^{V-1} (1 - Q^{j-m})$$

be the rank-deficiency probability of a uniform $m \times V$ matrix. If every geometric oil direction is rank deficient, then in particular all O standard oil-coordinate directions are rank deficient. Their derivative matrices are independent in the public-random expansion model. Hence

$$\Pr[F \in \mathcal{B}_{\text{oil}}] \leq \eta_{\text{oil}} := \tau_{m,V}^{O}. \tag{6}$$

Let $e = m - V$ and let $T_F : K^m \to \mathcal{U}$ be the coefficient map from the proof of Theorem A.10. The event $F \in \mathcal{B}_{\text{span}}$ is exactly $\dim \ker T_F \geq e$. For each fixed e-space $W \subset K^m$, the condition $W \subseteq \ker T_F$ imposes eD_{form} independent linear conditions on the uniform map T_F. A union bound over the Grassmannian therefore gives

$$\Pr[F \in \mathcal{B}_{\text{span}}] \leq \eta_{\text{span}} := \begin{bmatrix} m \\ e \end{bmatrix}_Q Q^{-eD_{\text{form}}}. \tag{7}$$

Thus the geometric proof requires only $\dim \mathrm{span}(F_1, \ldots, F_m) > V$; full linear independence is unnecessary.

Combining Theorem A.10 with (5), (6), and (7) gives the key theorem.

Theorem A.14 (Almost-all-key joint regularizability)**.** In the public-random expansion model,

$$\Pr[\Theta_F \equiv 0] \le \eta_{\mathrm{off}} + \eta_{\mathrm{oil}} + \eta_{\mathrm{span}}. \tag{8}$$

For the Round-2 parameter sets:

Level	c_{off}	$\log_2 \eta_{\mathrm{off}}$	$\log_2 \eta_{\mathrm{oil}}$	$\log_2 \Pr[\Theta_F \equiv 0]$
I	39	-364.564	-1132.167	< -364.563
III	55	-457.920	-1635.352	< -457.919
V	109	-1038.067	-2935.248	< -1038.066

The low-span term is below 2^{-94850} already at Level I and is invisible at the displayed precision.

Proof. By Theorem A.10, the event $\Theta_F \equiv 0$ is contained in $\mathcal{B}_{\mathrm{oil}} \cup \mathcal{B}_{\mathrm{off}} \cup \mathcal{B}_{\mathrm{span}}$. Apply (5), (6), (7), and the union bound. The displayed values are direct evaluations of those formulas, rounded outward at the shown precision. □

B Finite-field solver interfaces and verified restart

We first specialize a finite-field theorem for locally closed sets to the directional system. This gives a recovery route whose correctness needs only the local Jacobian condition of Section 6. We then return to the faster projective route and its near-quadratic solver. In both cases, every accepted key is checked against the unchanged public map.

B.1 Why the local Jacobian condition suffices

Proof of Lemma 6.1. Let $\mathfrak{p}$ be a prime of S_h containing the first s equations. Because h is a unit in S_h, the full Jacobian matrix is invertible modulo $\mathfrak{p}$, so its first s rows are linearly independent. The Jacobian criterion over a perfect field shows that the local zero set of $f_1, \dots, f_s$ is smooth of codimension s at every such prime. Hence the localized quotient is reduced and every minimal prime has height s. The ring S_h is regular and therefore Cohen–Macaulay. An ideal generated by s elements and of height s is a complete intersection, so the generators form a regular sequence. Smoothness gives radicality for every prefix. □

B.2 A simple product start over any sufficiently large finite field

Lemma B.1 (Vandermonde product start)**.** Let k' contain $2V$ distinct elements $a_1, \dots, a_{2V}$. For $a \in k'$, put

$$L_a(X) := X_1 + aX_2 + \cdots + a^{V-1}X_V + a^V, \tag{9}$$

and, for $1 \le i \le V$, set

$$g_i(X) := L_{a_{2i-1}}(X) L_{a_{2i}}(X). \tag{10}$$

Then $g = (g_1, \dots, g_V)$ has exactly $C = 2^V$ affine roots over k'. Every root is simple. A zero-dimensional parametrization of all roots can be constructed in $O(V^2 C)$ operations in k'.

Proof. Choose one node $b_i \in \{a_{2i-1}, a_{2i}\}$ from each pair. The equations $L_{b_1} = \cdots = L_{b_V} = 0$ have coefficient matrix

$$\begin{pmatrix} 1 & b_1 & \cdots & b_1^{V-1} \\ \vdots & \vdots & & \vdots \\ 1 & b_V & \cdots & b_V^{V-1} \end{pmatrix},$$

which is Vandermonde and invertible. Hence the choice gives one solution.

At that solution, the monic polynomial

$$T^V + X_V T^{V-1} + \cdots + X_2 T + X_1$$

has $b_1, \ldots, b_V$ as its roots. It therefore vanishes at none of the V unchosen nodes. Consequently the solution satisfies exactly one factor in each product g_i, and different choices give different solutions. Conversely, any common zero of the products must choose at least one factor from each pair. It cannot satisfy both factors in one pair, because that would give at least $V + 1$ distinct roots of the displayed monic polynomial. Thus there are exactly 2^V roots.

At a root, row i of the Jacobian of g is the Vandermonde row belonging to the chosen node, multiplied by the nonzero value of the unchosen factor. The Jacobian is therefore an invertible Vandermonde matrix times an invertible diagonal matrix. Enumerating the 2^V choices and forming the corresponding degree-V polynomials gives the stated $O(V^2 C)$ bound. $\square$

This construction uses no division by an integer and no condition on the characteristic beyond the existence of $2V$ distinct field elements. For the QR-UOV working extensions below, this condition is trivial.

B.3 A full random separator

A primitive linear form must separate finitely many branch values and avoid finitely many leading-coefficient vectors. The published proof draws from a short integer-indexed family; that is the source of its second characteristic bound. Sampling the entire dual space is simpler over a finite extension.

Lemma B.2 (Separator over a finite extension)**.** Let A be a domain containing the finite field k', and let $z_1, \ldots, z_s \in A^V$ be nonzero. For a uniform $\lambda = (\lambda_1, \ldots, \lambda_V) \leftarrow (k')^V$,

$$\Pr[\lambda \cdot z_j = 0 \text{ for some } j] \leq \frac{s}{|k'|}. \tag{11}$$

In particular, if $s \leq C^2$, the separator succeeds with probability at least $1 - C^2/|k'|$.

Proof. Fix a nonzero vector z. Choose an index h with $z_h \neq 0$. After the other $V - 1$ coefficients of λ are fixed, at most one value of $\lambda_h \in k'$ can make $\lambda \cdot z = 0$: two such values would give a nonzero scalar in k' annihilating the nonzero domain element z_h. Thus one vector is killed with probability at most $1/|k'|$. A union bound proves the claim. $\square$

Safey El Din and Schost identify at most C^2 vectors that a well-separating form must avoid: differences between start roots, differences between relevant target specializations, and leading-coefficient vectors of branches [27, Lem. 14 and Rem. 15]. Lemma B.2 therefore replaces their integer family without changing the success criterion.

B.4 The lifting identities do not require large characteristic

For completeness, we isolate the two local identities used by precision doubling. They work over a commutative ring and do not divide by factorials or by the characteristic.

Lemma B.3 (Newton–Hensel precision doubling)**.** Let R be a commutative ring, let $I \subset R$ be an ideal, and let $F \in R[X_1, \ldots, X_V]^V$. Suppose

$$F(x) \equiv 0 \pmod{I^m}$$

and that $J_F(x)$ is invertible modulo I^m. Put

$$x^+ := x - J_F(x)^{-1}F(x) \pmod{I^{2m}}. \tag{12}$$

Then $F(x^+) \equiv 0 \pmod{I^{2m}}$.

Proof. The correction $h = x^+ - x$ lies in I^m. Polynomial Taylor expansion gives

$$F(x+h) = F(x) + J_F(x)h + R_2(x,h),$$

where every monomial in R_2 contains at least two coordinates of h. Hence $R_2(x,h) \in I^{2m}$. The first two terms cancel by (12). No denominator is used. □

Lemma B.4 (First-order change of primitive parameter)**.** Work modulo T^{2m}. Let $Q(T,U)$ be monic in U, let $X(T,U) \in (R[T,U]/(T^{2m},Q))^V$, and let λ be linear. Suppose

$$e(T,U) := \lambda(X(T,U)) - U \in T^m R[T,U]/(T^{2m},Q).$$

Define

$$X^* := X - (\partial_U X)e, \qquad Q^* := Q - (\partial_U Q)e. \tag{13}$$

Then the change of parameter $U_{\mathrm{old}} = U - e(T,U)$ represents the same lifted branches modulo T^{2m}, and

$$\lambda(X^*) \equiv U \pmod{(T^{2m}, Q^*)}. \tag{14}$$

Proof. Because $e \in T^m$, substitution of $U - e$ into any polynomial is, modulo T^{2m}, its first-order Taylor expansion. This gives exactly (13). Moreover

$$\lambda(X^*) - U = e - (\partial_U \lambda(X))e = -(\partial_U e)e.$$

Differentiation in U does not change T-adic order, so $\partial_U e \in T^m$ and the product lies in T^{2m}. Again, the argument uses no integer denominator. □

B.5 Proof crosswalk for the finite-characteristic solver

Proposition B.5 (Where the published characteristic bound is used)**.** For a square system of quadrics over a finite perfect field, replace the start construction in Safey El Din–Schost Lemma 7 by Lemma B.1, and replace the separator family in their Lemma 14 by Lemma B.2. The remaining correctness proof of their NonsingularSolutions algorithm applies over the extension field without a further lower bound on the characteristic.

Proof. We follow the proof in the order in which its ingredients are used.

Start fiber. The original lower bound $\operatorname{char} k \geq \sum_i d_i$ ensures that its integer-indexed affine forms are distinct and that the product start has the full multihomogeneous root count with invertible Jacobian. Lemma B.1 supplies the same three facts directly: degree two, exactly C explicit roots, and a simple start fiber.

Homotopy curve. The homotopy is

$$H(T,X) = (1-T)g(X) + Tf(X).$$

The published construction saturates by the X-Jacobian determinant. Lazard's lemma and the multihomogeneous degree bound then give a one-dimensional curve of degree at most C_0. These arguments require the simple start just established, not an ordering of integers inside the field.

Target specialization. The target system can have fewer than C isolated nonsingular roots and may have singular or positive-dimensional components. The published proof handles this by generalized power series, explicitly because ordinary Puiseux series are not sufficient in arbitrary characteristic [27, Sec. 3.4]. Its valuation and specialization lemmas therefore already cover the finite-field setting.

Precision doubling and primitive normalization. The global lifting step is Newton–Hensel lifting of a geometric resolution. Its local identities are Lemmas B.3 and B.4; interpolation and polynomial arithmetic use inverses guaranteed by the simple fiber and separating form. No factorial or characteristic-dependent scalar is introduced.

Separator and cleaning. The published well-separating condition is avoidance of at most C^2 nonzero vectors. Lemma B.2 supplies it with probability at least $1 - C^2/|k'|$. Finally, the published CLEAN call removes every point where the target Jacobian vanishes. As noted in its Remark 15, every output is therefore a subset of the nonsingular target roots; a bad separator may omit roots or cause failure, but cannot introduce an invalid point.

These replacements cover the two uses of the integer-indexed finite families. All other steps are field algebra, generalized-power-series specialization, or Jacobian cleaning over the perfect extension field. □

B.6 Locally closed finite-field solving

The following proposition records the exact external interface used below. It is a specialization of Theorem 12 of Giménez et al. [9].

Proposition B.6 (Locally closed finite-field solver)**.** Let $F_1, \dots, F_r, H \in \mathbb{F}_{\mathfrak{q}}[X_1, \dots, X_n]$ have degree at most d. Suppose that, in the localization $\mathbb{F}_{\mathfrak{q}}[X]_H$,

1. $F_1, \dots, F_r$ form a regular sequence; and
2. $(F_1, \dots, F_s)$ is radical for every $1 \le s \le r$.

Let V_s be the Zariski closure of $\{F_1 = \cdots = F_s = 0,\ H \neq 0\}$, put $\delta_s = \deg V_s$, and let $\delta = \max_s \delta_s$. If the input has division-free straight-line-program length L, $0 < \epsilon < 1/4$, and

$$\mathfrak{q} > 2\epsilon^{-1} n^2 r d \delta^3, \tag{15}$$

then a probabilistic algorithm computes a Kronecker representation of a lifting fiber of V_r with probability at least $1 - 2\epsilon$ and bit complexity

$$\widetilde{O}\Big((n^4 + r(L + n^2 + r^4)) d\delta^2 \log_2 \mathfrak{q} \Big) . \tag{16}$$

When $r = n$, the lifting fiber is the entire zero-dimensional locally closed set. The solver is Monte Carlo: its paper does not provide an efficient certificate that a returned representation is complete.

Proof. The two displayed algebraic conditions are hypotheses (H1)–(H2) of the cited theorem. Equation (15), success probability $1 - 2\epsilon$, and (16) are its finite-field statement. The source explicitly notes that its algorithms do not appear to be Las Vegas because output correctness cannot in general be verified at comparable cost. For $r = n$, no free projection coordinates remain, so a lifting fiber is the zero-dimensional set itself. The source treats $\delta > d$ as without loss of generality; if $\delta \le d$, replacing it by the safe upper bound $d + 1$ only enlarges the field and cost bounds. □

Corollary B.7 (Graph-local specialization)**.** Assume $\operatorname{rank} \mathcal{B}(c) = V$ and $R\mathcal{B}(c)$ is invertible. Apply Proposition B.6 to the V affine quadrics $f = Rq_c$ and the localization polynomial $h = \det J_f$. Then

$$n = r = V, \qquad d = V, \qquad \delta \le D := 2^V .$$

The input has straight-line-program length

$$L_{\mathrm{loc}} = O(L_{\mathrm{core}} + V^4), \qquad L_{\mathrm{core}} := M_V + 2V(M_V + V + 1), \quad M_V := \frac{V(V+1)}{2}. \tag{17}$$

Over any extension $L = \mathbb{F}_{Q^s}$ satisfying

$$Q^s > 2\epsilon^{-1}V^4 8^V, \tag{18}$$

the solver contains the planted root Gc with probability at least $1 - 2\epsilon$ and has bit complexity

$$\widetilde{O}\Big((V^4 + V(L_{\mathrm{loc}} + V^2 + V^4))V4^V \log_2(Q^s)\Big). \tag{19}$$

Proof. Corollary 6.2 gives the localized regularity and radicality hypotheses and places Gc in the open set. Bézout bounds every intermediate degree by $2^s \le 2^V$. The maximum input degree is V because h has degree V. The dense quadratic core has the shared-monomial circuit in (17); Baur–Strassen differentiation followed by a division-free Samuelson–Berkowitz determinant circuit adds $O(V^4)$ operations. Substitution into (15)–(16) proves the result. □

Corollary B.8 (Smooth-projective Gim'enez cross-check)**.** On a smooth projective core, choose a hyperplane at infinity and dehomogenize in the complementary affine chart. Conditional on the hyperplane avoiding all $D = 2^V$ projective roots, apply Proposition B.6 with $H = 1$. Then

$$n = r = V, \qquad d = 2, \qquad \delta \le D,$$

and, over $\mathbb{F}_{Q^s}$ with

$$Q^s > 4\epsilon^{-1}V^3 8^V, \tag{20}$$

the direct displayed complexity is

$$\widetilde{O}\Big(2(V^4 + V(L_{\mathrm{core}} + V^2 + V^4))4^V \log_2(Q^s)\Big). \tag{21}$$

The conditional solver success probability is at least $1-2\epsilon$. If the hyperplane is sampled uniformly over $\mathbb{F}_{Q^s}$, the unconditional success probability of the chart-plus-solver step is at least

$$1 - 2\epsilon - \frac{D}{Q^s}. \tag{22}$$

Proof. Smoothness of the full projective zero scheme implies regularity and geometric radicality of every prefix by Lemma A.4. A hyperplane avoiding the finite zero set exists, and dehomogenization in its complementary affine chart preserves those properties. Substitute $d = 2$ and $\delta \le 2^V$ into Proposition B.6. A uniformly random hyperplane contains any fixed projective point with probability at most $1/Q^s$; a union bound over the D roots gives the additional D/Q^s term in (22). □

Why Monte Carlo output is sufficient here. A successful call to Proposition B.6 contains the planted nonsingular root. A failed call may omit it or return unusable data. The attack parses every returned representation, retains only actual roots of all original equations, completes and verifies the graph, checks the inverse pull-back and public-map factorization, and finally checks an invertible oil system. If no candidate passes, it retries with fresh coins. Thus the external solver's lack of a completeness certificate can delay termination but cannot create a false accepted key.

B.7 Fast projective route: field size, failure probability, and cost

Put

$$D := 2^V, \qquad L_s := \mathbb{F}_{Q^s}, \qquad q_s := |L_s|.$$

The finite-field theorem of van der Hoeven and Lecerf has an exponent parameter and a field-size requirement. They must be chosen together. We use $\varepsilon > 0$ for the desired final exponent and set the theorem's internal parameter to $\eta = \varepsilon/3$. For V quadrics, its field-size threshold is

$$B_V(\eta) := \max\Big\{\max\Big\{2, (1+\eta)(8/\eta+1)V2^{\eta/8}\Big\}(2^V-1), 100\,8^V\Big\}. \tag{23}$$

Define the smallest admissible extension degree and a one-degree safety choice by

$$s_\varepsilon^0 := \min\{s \geq 1 : Q^s > B_V(\varepsilon/3)\}, \qquad s_\varepsilon^+ := s_\varepsilon^0 + 1. \tag{24}$$

The first choice minimizes field size. The second divides all finite-invocation failure bounds by the large factor $Q = 2{,}048{,}383$.

For any $s \geq s_\varepsilon^0$, sample

$$M \leftarrow L_s^{(V+O)\times(V+1)}, \qquad R \leftarrow L_s^{V\times m}$$

and form the restricted square core. For every fixed key with $\Theta_F \not\equiv 0$, Schwartz–Zippel gives

$$\beta_{V,s} := \Pr[\Theta_F(M,R) = 0] \leq \frac{E_V^{\text{joint}}}{Q^s}, \qquad E_V^{\text{joint}} := 3V(V+1)2^{V-1}. \tag{25}$$

The proof of the finite solver theorem bounds its three internal bad choices by

$$\alpha_{V,s} := \frac{5D^2 + 26D^3}{Q^s}. \tag{26}$$

These are bounds for one finite invocation. A denominator failure, degree mismatch, or failed final check returns fail; the algorithm never runs an unproved internal restart loop on an irregular outer draw.

For the primary choice $\varepsilon = 0.3$, the field threshold is governed by $100\,8^V$. The minimum and safety choices are:

Level	$s_{0.3}^0$	success at $s_{0.3}^0$	$s_{0.3}^+$	success at $s_{0.3}^+$
I	8	≥ 0.992337625602	9	≥ 0.999999996259
III	12	≥ 0.999997944684	13	≥ 0.999999999998
V	15	≥ 0.927725586161	16	≥ 0.999999964716

Here success means that the outer core is smooth and the solver's internal choices are good, conditional on $\Theta_F \not\equiv 0$; it is at least $(1-\beta_{V,s})(1-\alpha_{V,s})$. At the safety fields, the base-two logarithms of $(\alpha_{V,s}, \beta_{V,s})$ are

$$(-27.994, -124.681), \qquad (-39.858, -183.459), \qquad (-24.756, -219.513).$$

Thus the minimum fields are adequate for expected-time verified restart, while the safety fields give a clean one-invocation theorem. The dependence on ε in (24) is essential: no fixed list of extension degrees works for every arbitrarily small fixed $\varepsilon > 0$.

B.8 Published near-quadratic Kronecker solver

Proposition B.9 (Fast smooth-core specialization)**.** Let $g_1, \ldots, g_V$ be homogeneous quadrics over a finite field $\mathbb{F}_{p^k}$. Assume that they form a regular sequence and every projective prefix is geometrically radical. Put $D_i = 2^i$ and $D = D_V$. Fix a rational $\varepsilon > 0$, set $\eta = \varepsilon/3$, and assume

$$p^k > B_V(\eta).$$

One finite invocation of Theorem 5.4 of van der Hoeven and Lecerf uses

$$\widetilde{O}\Big(D\, 2^{(1+\varepsilon)(V-1)} k \log_2 p\Big) \tag{27}$$

bit operations and fails because of its random coordinate, intersection, or evaluation choices with probability at most

$$\frac{5D^2 + 26D^3}{p^k}. \tag{28}$$

Outside those bad choices it returns a complete Kronecker parametrization. The checks in Proposition B.11, including the requirement that the separator have degree D, certify completeness for this task. Repeating checked finite invocations on an input satisfying the hypotheses gives the Las Vegas expected-time form of Corollary 5.5.

For $K = \mathbb{F}_{127^3}$ and any $s \geq s_\varepsilon^0$, the explicit finite-invocation cost over L_s is obtained from (27) with $k = 3s$ and $p = 127$. Since $s = O(V) = O(\log D)$ for the choices above, this field-degree factor is absorbed by soft-O notation. Including field construction, casting, public slicing, output recombination, solving, and verification, the full expected projective route has scale

$$T_{\varepsilon,V} := \widetilde{O}\Big(2^V 2^{(1+\varepsilon)(V-1)} 3 \log_2 127\Big). \tag{29}$$

Proof. The first statement is Theorem 5.4 specialized to the degree sequence $(2, \ldots, 2)$, with internal parameter $\eta = \varepsilon/3$. Its complete field condition is (23); its complexity exponent $1 + 3\eta$ becomes $1 + \varepsilon$. The three bad-choice bounds in the proof sum to (28).

A smooth zero-dimensional projective core satisfies the required regularity and prefix-radicality assumptions by Lemma A.4. Corollary 5.5 of the source constructs a suitable extension, casts the input, verifies the returned parametrization, and obtains the same expected soft-O scale. Sampling M and R, restricting the dense quadrics, and forming the output combinations costs only polynomially many operations in the working field and is lower order than the degree-dependent term. □

Corollary B.10 (One checked projective invocation)**.** Fix $\varepsilon > 0$, choose any $s \geq s_\varepsilon^0$, and fix a key with $\Theta_F \not\equiv 0$. Run one outer draw over L_s, one finite invocation from Proposition B.9, and all checks from Proposition B.11; return `fail` if any check rejects. The invocation has soft-O cost $T_{\varepsilon,V}$ and succeeds with probability at least

$$(1 - \beta_{V,s})(1 - \alpha_{V,s}). \tag{30}$$

For $\varepsilon = 0.3$, the minimum-field and safety-field probabilities are those in the preceding table.

Proof. The outer discriminant fails with probability at most $\beta_{V,s}$. Conditional on a smooth core, Proposition B.9 fails internally with probability at most $\alpha_{V,s}$. The two groups of coins are fresh. Exact checking changes malformed or incomplete output into `fail`, never into a false accepted result. □

B.9 A classical arithmetic cross-check

Let

$$M_2 := \binom{V+2}{2}, \qquad L_{\mathrm{quad}} := M_2 + V(2M_2 - 1). \tag{31}$$

The first term computes every degree-two monomial in the $V+1$ homogeneous variables once. Each of the V dense quadrics then uses M_2 scalar multiplications and $M_2 - 1$ additions. Thus $L_{\mathrm{quad}} = \Theta(V^3)$ is an explicit division-free circuit for the full square core.

Giusti–Lecerf–Salvy give the older arithmetic bound

$$L_{\mathrm{quad}}\, \widetilde{O}(D^2) \tag{32}$$

field operations under the same regularity conditions [8, 7]. We retain (32) only as an epsilon-free growth-rate cross-check. We do *not* transfer the explicit finite-field failure constants of Theorem 5.4 to this older bound: the literature cited here does not state that combined theorem. Accordingly, the SLP line below has no proved one-invocation failure probability and is not used for the universal-forgery corollary.

Proposition B.11 (Task-specific resolution checks)**.** Let $f = (f_1, \ldots, f_V)$ be affine quadrics over a finite field L, and put $D = 2^V$. Suppose a proposed geometric resolution consists of a monic polynomial $m(T)$, coordinate polynomials $v_1(T), \ldots, v_V(T)$, and a linear form u. Reject it unless every v_j has degree below $\deg m$ and

1. $1 \le \deg m \le D$ and m is squarefree;
2. $u(v_1(T), \ldots, v_V(T)) \equiv T \pmod{m}$;
3. $f_i(v_1(T), \ldots, v_V(T)) \equiv 0 \pmod{m}$ for every i;
4. $\gcd(m(T), \det J_f(v_1(T), \ldots, v_V(T))) = 1$.

If the solver returns standard numerators w_j with denominator m', first verify $\gcd(m, m') = 1$ and set $v_j = w_j(m')^{-1} \bmod m$. Passing these checks certifies that the represented points are distinct simple solutions of the square core. When this proposition is used with a smooth zero-dimensional complete intersection of V quadrics, requiring in addition $\deg m = D$ certifies completeness by Bézout. The checks cost $\widetilde{O}(\mathrm{poly}(V) D \log |L|)$ Boolean gates.

Proof. Squarefreeness gives distinct roots τ of m over $\overline{L}$. The primitive-form identity makes the represented points $P_\tau = (v_1(\tau), \ldots, v_V(\tau))$ distinct. The equation checks put them in $V(f)$, and the Jacobian gcd makes each one simple. Evaluate the quadrics and Jacobian in $L[T]/(m)$; compute the determinant there with a division-free algorithm because this quotient need not be a field. Polynomial arithmetic, gcds, and the $V \times V$ determinant have the stated cost. □

Why the extra checks remain useful. The published Las Vegas solver already verifies its complete parametrization. The task-specific checks make the cryptanalytic interface self-contained and protect against malformed implementations, truncated outputs, and conversion errors. On every branch, final acceptance additionally requires all unrecombined equations, exact public-map factorization, and an invertible oil system. No malformed solver transcript can create a false accepted key.

B.10 Root selection and oil-space recovery

Lemma B.12 (Recovery from the projective oil point)**.** Let L/K be a finite extension. Suppose $\Theta_F(M, R) \neq 0$, and let $[t^\star] \in \mathbb{P}^V(L)$ be the unique point for which $x^\star = Mt^\star$ lies in $\mathbb{P}(\mathcal{O})_L$. Then

$$\mathrm{rank}(P_1 x^\star \ \cdots \ P_m x^\star) = V, \qquad \ker(P_1 x^\star \ \cdots \ P_m x^\star)^\mathsf{T} = \mathcal{O}_L.$$

Row reduction recovers the public graph of the oil space. For the planted point that graph is defined over K and, after the specification's inverse pull-back, yields the planted expanded UOV decomposition.

Proof. The full-rank matrix M defines a projective V-plane over L. Its vector-space intersection with the O-dimensional oil space is nonzero by dimension count. If that intersection had dimension at least two, all restricted quadrics would vanish on a positive-dimensional projective linear space, contradicting smooth zero-dimensionality. Thus it is one L-rational point.

Smoothness of the square core gives derivative rank at least V there. Total isotropy puts the O-dimensional oil space inside the ambient common derivative kernel, so the rank is at most $(V+O)-O=V$. Equality holds, and Lemma 5.5 identifies the kernel with $\mathcal{O}_L$. The remaining statements follow from K-descent and Proposition 3.4. □

Lemma B.13 (Separator descent)**.** Let a reduced zero-dimensional scheme over $L=\mathbb{F}_{\mathfrak{q}}$ have a Kronecker representation whose primitive linear form u separates its geometric points. A geometric point P is L-rational if and only if $u(P)\in L$.

Proof. The forward implication is immediate. Conversely, if $u(P)\in L$, then

$$u(P^{(\mathfrak{q})})=u(P)^{\mathfrak{q}}=u(P).$$

Because u is defined over L and separates the geometric points, Frobenius must fix P. □

Proposition B.14 (Filter all original equations before factoring)**.** Let a consistency-checked resolution over L_s have squarefree separator polynomial $m_u(T)$ and coordinate polynomials $v(T)$. Let $\widetilde{q}_1,\dots,\widetilde{q}_m$ be the dehomogenized original restricted quadrics in the same affine chart, and define

$$a(T):=\gcd\bigl(m_u(T),\widetilde{q}_1(v(T)),\dots,\widetilde{q}_m(v(T))\bigr). \tag{33}$$

Then the roots of a are exactly the separator values of represented points satisfying all m original equations. Moreover,

$$a_{L_s}(T):=\gcd\bigl(a(T),T^{q_s}-T\bigr) \tag{34}$$

has exactly the separator values of those common points that are L_s-rational. Computing a and a_{L_s} costs $\widetilde{O}(D\ \mathrm{poly}(V,m,\log|L_s|))$ Boolean gates. Only a_{L_s}, whose degree may be much smaller than D, must be factored.

Proof. Work over an algebraic closure of L_s. Because m_u is squarefree, it is the product of the distinct factors $T-\tau$ indexed by the represented points. For each root τ,

$$\widetilde{q}_i(v(T)) \bmod m_u\,\big|_{T=\tau}=\widetilde{q}_i(v(\tau)).$$

Thus $T-\tau$ divides (33) exactly when every original restricted equation vanishes at the represented point. The second claim follows from Lemma B.13 and the identity $z^{q_s}=z$ for $z\in L_s$. Dense quadratic evaluation in $L_s[T]/(m_u)$, modular exponentiation, and polynomial gcd computation at degree at most D give the stated cost. Since $\log|L_s|=O(V)$ for the selected fields, it is lower order than the degree-squared solver bound. □

Check the resolution, prefilter, and extract rational roots. Convert the solver output to canonical coordinate polynomials and run the consistency checks of Proposition B.11. On failure, discard the invocation. On a successful Las Vegas solver call, the returned separator polynomial $m_u(T) \in L_s[T]$ represents the full affine zero scheme and contains the planted oil point. The additional checks reject malformed or incomplete implementation outputs.

Apply Proposition B.14: evaluate all m original restricted equations in $L_s[T]/(m_u)$, compute $a(T)$, and then compute $a_{L_s}(T)$. Factor only a_{L_s} over L_s. A Las Vegas finite-field factorization algorithm terminates with probability one and has expected subquadratic dependence on the input degree, uniformly for inputs of degree at most D; its expected contribution is therefore below the solver's near-quadratic-in-D scale [22]. Reconstruct the corresponding affine points, undo the coordinate changes, and recheck all m original equations as a final exact verification.

Certify the decomposition. For each survivor x, compute the common derivative kernel. Require an O-dimensional graph defined over K, total isotropy, and every identity (3). Apply the specification's inverse pull-back and verify the base-field factorization against the original public map. An output is called an *exact candidate* only after all these checks pass.

Certify signing or restart. Deduplicate exact candidates by their K-rational graph. For each remaining candidate, first test the V basis assignments from Proposition 3.7. If they all fail, scan the remaining nonzero combinations on the designated pair lines. Return the first candidate with an invertible oil matrix, together with that matrix's inverse. If no candidate passes, discard the trial and sample a fresh slice. On a correct solver branch the candidate list contains the planted decomposition, so outside the event in (15) this scan succeeds deterministically. At most

$$D(V + \lfloor V/2 \rfloor (q-1))$$

oil matrices are tested per trial, below the solver's near-quadratic-in-D scale. In the public-random model, the basis-first part succeeds after fewer than 1.008 tests on average.

Let $\mathsf{A}_{\varepsilon,V,s}$ denote the event that the full trial accepts a publicly verified signing decomposition and inverse oil matrix. On a key outside $\mathcal{E}_{\text{sign}}$, a draw with $\Theta_F(M,R) \neq 0$ yields the planted exact candidate whenever the finite solver invocation succeeds, after which the deterministic signing scan succeeds. The outer and internal coins are fresh, so

$$\Pr[\mathsf{A}_{\varepsilon,V,s}] \geq (1-\beta_{V,s})(1-\alpha_{V,s}). \tag{35}$$

This one-sided implication is the only interface needed between the solver guarantee and the verified attack. On irregular inputs or bad internal choices, a failed check returns fail; it never creates a false accepted key.

Lemma B.15 (Verified-restart accounting)**.** Suppose each fresh trial, conditional on any history of earlier failures, accepts with probability at least $p > 0$. Then the number N of trials satisfies

$$\Pr[N > t] \leq (1-p)^t, \qquad \mathbb{E}[N] \leq p^{-1}.$$

If the current trial uses fresh coins and has expected work at most T, uniformly over prior histories, then the expected total work is at most T/p.

Proof. The tail bound follows inductively from the conditional success lower bound, and summing the tail probabilities gives $\mathbb{E}[N] \leq 1/p$. For the work bound, the event that trial j is reached depends only on earlier trials. Fresh current-trial coins therefore give

$$\mathbb{E}\Big[\mathbf{1}_{\{N \geq j\}} W_j\Big] \leq T \Pr[N \geq j].$$

Summing over j and applying the trial bound proves the claim. □

B.11 Main recovery theorem

Use $T_{\varepsilon,V}$ from (29). For $0 < \epsilon < 1/4$ and an extension degree s satisfying (18), write

$$T_{\mathrm{loc}}(\epsilon, s) := \widetilde{O}\Big((V^4 + V(L_{\mathrm{loc}} + V^2 + V^4))V4^V \log_2(Q^s)\Big) .$$

Theorem B.16 (Key-by-key recovery from one graph direction)**.** Let an expanded QR-UOV key over K have a valid planted decomposition. Suppose that at least one maximal minor of $\mathcal{B}(c)$ is a nonzero polynomial in c. Fix $0 < \epsilon < 1/4$ and choose s so that (18) holds. Put

$$p_{\mathrm{dir}} := \left(1 - \frac{V}{Q}\right)\left(1 - \frac{V(m-V)}{Q}\right)(1 - 2\epsilon).$$

Then the following verified attacks are correct for this fixed key.

1. A randomized graph-direction trial finds an exactly verified expanded UOV decomposition with probability at least p_{dir}. Repeating fresh trials terminates almost surely, uses at most p_{dir}^{-1} trials on average, and has expected work

   $$O(T_{\mathrm{loc}}(\epsilon, s)/p_{\mathrm{dir}}) . \tag{36}$$

   Every accepted output is correct.

2. If the determinant polynomial Δ of the planted decomposition is nonzero, testing one fresh uniform scalar-block vinegar assignment in each trial returns a signing certificate with expected trial count at most

   $$\big(p_{\mathrm{dir}}(1 - m/q)\big)^{-1}. \tag{37}$$

3. If one planted basis matrix is invertible, the deterministic basis-first signing scan succeeds on every successful recovery trial. Under the designated-pair condition of Proposition 3.7, the full basis-and-line scan does so after at most $V + \lfloor V/2 \rfloor(q-1)$ oil-matrix tests per candidate. In either case the expected trial and work bounds are those of item 1.

4. If a coordinate direction e_j satisfies $\operatorname{rank}\mathcal{B}(e_j) = V$, all outer attack choices can be made deterministic: scan the $O\left[V(m-V)+1\right]$ direction/condenser pairs of Corollary D.3, and repeat the finitely many probabilistic solver calls in rounds. This still terminates almost surely and adds only a polynomial factor. The shorter lists of Corollary D.4 give category-scale failure bounds in the public-random model.

The theorem uses only a local nonsingularity condition at the planted root. It does not require the projective discriminant, the off-oil incidence argument, a key-generation distribution, or uniqueness of the recovered decomposition.

Proof. **A good trial contains the planted root.** By Proposition 5.4, a random c makes $\mathcal{B}(c)$ full rank with probability at least $1 - V/Q$. Conditional on that event, a random t makes $\mathsf{C}(t)\mathcal{B}(c)$ invertible with probability at least $1 - V(m-V)/Q$. The planted root then lies in the localized smooth set by Corollary 6.2. By Corollary B.7, the solver returns a representation containing it with probability at least $1 - 2\epsilon$.

One planted root gives the whole hidden graph. For every returned root x, solve the public systems (39). At $x = Gc$, Proposition 5.2 returns every column of G. The attack then checks all original direction equations, rank conditions, graph identities, inverse pull-back identities, and public-map factorizations. Any root that does not yield a valid graph is discarded.

Restart and signing. A fresh trial therefore reaches the planted expanded decomposition with probability at least p_{dir}. Apply Lemma B.15. The randomized and deterministic signing claims follow from Proposition 3.7. Filtering and completing at most D roots costs $\widetilde{O}(D\,\mathrm{poly}(V, O, m, \log|L|))$, below (19). Exact verification prevents false acceptance even when the internal Monte Carlo solver fails.

For item 4, Corollary D.3 guarantees that a regular coordinate direction has a condenser value with nonsingular planted Jacobian. The deterministic scan reaches it. Repeating all finitely many solver calls in rounds gives a fixed positive success probability per round and hence almost-sure termination. □

Corollary B.17 (Public-random density and compact deterministic certificates)**.** In the public-random expansion model, the probability that all O coordinate directions are rank deficient is exactly $\tau_{m,V}(Q)^O$, whose base-two logarithms are

$$-1132.166907, \qquad -1635.352198, \qquad -2935.247544.$$

Hence, except with probability at most

$$\eta_{\mathrm{local}} := \tau_{m,V}(Q)^O + \eta_{\mathrm{sign}}, \tag{38}$$

the full deterministic direction list and deterministic signing fallback recover a universal signing certificate.

A much shorter fixed attack already has category-scale coverage. Use the first $3, 4, 4$ coordinate directions and their condenser lists, together with the first $22, 31, 40$ basis signing assignments, at Levels I, III, and V. These attacks solve only $315, 612, 1228$ square systems and test at most $22, 31, 40$ oil matrices per exact candidate. Their failure probabilities are at most

$$\tau_{m,V}(Q)^{r_{\mathrm{dir}}} + \sigma_m^{b_{\mathrm{sign}}},$$

with base-two logarithms below

$$-153.502, \qquad -216.298, \qquad -279.095. \tag{39}$$

No independence between the recovery and signing failures is needed for this union bound.

Proof. The all-direction statement is (42), followed by Proposition 3.7 and the union bound. For the compact statement, combine Corollary D.4 with the fixed-prefix signing bound (14). On the complement, the deterministic outer list contains a square system with the planted point nonsingular, and the fixed signing list contains an invertible planted oil matrix. Round-robin solver repetition and exact verification then give the claimed universal certificate. □

Theorem B.18 (Fast projective key-by-key recovery)**.** Let an expanded QR-UOV key over K have a valid planted expanded decomposition, and assume that the public joint slice-and-output discriminant $\Theta_F(M, R)$ is not the zero polynomial. Fix a rational $\varepsilon > 0$, choose any $s \geq s_\varepsilon^0$, and put

$$p_{\varepsilon,V,s} := (1 - \beta_{V,s})(1 - \alpha_{V,s}).$$

Then:

1. A decomposition-only verified-restart algorithm terminates with probability one and returns an exactly verified expanded UOV decomposition. Every accepted output is correct, the expected number of outer trials is at most $p_{\varepsilon,V,s}^{-1}$, and the expected work is

$$O\left(\frac{T_{\varepsilon,V}}{p_{\varepsilon,V,s}}\right). \tag{40}$$

For $\varepsilon = 0.3$, the minimum fields are $s = 8, 12, 15$. Their expected-trial bounds are below

$$1.00773, \qquad 1.000003, \qquad 1.07791.$$

The safety fields $s = 9, 13, 16$ give the near-one single-invocation probabilities in Corollary B.10.

2. Suppose the planted determinant polynomial Δ from Proposition 3.7 is nonzero. If each outer trial tests one fresh uniform scalar-block vinegar assignment for every exact candidate, verified restart returns an equivalent signing key and inverse oil matrix. The expected number of outer trials is at most
$$\frac{1}{p_{\varepsilon,V,s}(1-m/q)},$$
and the expected work is
$$O\left(\frac{T_{\varepsilon,V}}{p_{\varepsilon,V,s}(1-m/q)}\right). \tag{41}$$
At most D oil matrices are tested per trial.

3. If one planted basis matrix M_j is invertible, the basis-first deterministic scan returns an equivalent signing key on every successful core invocation and tests at most DV oil matrices per outer trial. Under the designated-pair condition of Proposition 3.7, the full deterministic fallback tests at most
$$D(V+\lfloor V/2\rfloor(q-1))$$
oil matrices per trial. In either case the expected trial count is at most $p_{\varepsilon,V,s}^{-1}$ and the expected work remains $O(T_{\varepsilon,V}/p_{\varepsilon,V,s})$.

These are statements about each fixed expanded key satisfying the displayed conditions. They do not depend on how the key was generated. Recovering the original seed or serialized private key is unnecessary.

Proof. **A regular draw yields a complete core resolution.** The discriminant hypothesis and (25) imply that a fresh (M,R) gives a smooth full core with probability at least $1-\beta_{V,s}$. Lemma A.4 supplies the solver hypotheses. Conditional on a smooth core, the finite invocation in Proposition B.9 returns and verifies the complete geometric resolution with probability at least $1-\alpha_{V,s}$. Every invocation has a finite operation schedule and returns fail if a denominator, degree, or verification check rejects. An irregular outer draw therefore cannot trap the attack inside an unjustified restart loop.

The planted decomposition appears among the exact candidates. On a regular draw, the quotient-algebra prefilter of Proposition B.14, rational-root extraction, and Lemma B.12 retain the planted oil point and recover the planted expanded decomposition. Exact all-equation checks, inverse pull-back, and public-map factorization reject every invalid candidate.

Signing certification. If $\Delta\not\equiv 0$, Proposition 3.7 gives success probability at least $1-m/q$ for a fresh uniform assignment on the planted candidate. If a basis matrix or designated pair witness exists, the corresponding deterministic scan succeeds on every regular draw.

Restart and auxiliary work. Apply Lemma B.15. Finite-field factorization has subquadratic expected dependence on degree, and all filtering, derivative-kernel, descent, pull-back, verification, and signing work costs $\widetilde{O}(D\,\mathrm{poly}(V,O,m,q,\log|L_s|))$. For every fixed $\varepsilon>0$ this is lower order than $T_{\varepsilon,V}$. Exact verification prevents false acceptance on every branch. $\square$

Corollary B.19 (Public-random density for the fast projective route)**.** In the public-random expansion model, the probability that either $\Theta_F\equiv 0$ or the planted determinant polynomial Δ is identically zero is below
$$2^{-364.563},\qquad 2^{-457.919},\qquad 2^{-1038.066}$$
at Levels I, III, and V. Every other key satisfies the randomized equivalent-key conclusion of Theorem B.18. In fact, outside the same bounded union of bad events, one designated pair span contains an invertible matrix, so the deterministic refinement applies.

Proof. The event $\Theta_F \equiv 0$ is bounded by Theorem A.14. If Δ is the zero polynomial, every designated pair span is singular, so this event is contained in $\mathcal{E}_{\text{sign}}$ and is bounded by Proposition 3.7. The union has probability at most

$$\eta_{\text{off}} + \eta_{\text{oil}} + \eta_{\text{span}} + \eta_{\text{sign}},$$

which gives the displayed numerical bounds. On the complement of this particular union, the designated-pair condition itself holds, so item 3 of Theorem B.18 applies. □

Corollary B.20 (One-invocation key-only universal forger)**.** In the public-random expansion model, set $\varepsilon = 0.3$, use the safety field $s = s_{0.3}^{+}$, run the checked projective invocation of Corollary B.10, and then run the basis-first scan and projective-line fallback of Proposition 3.7. The adversary uses no signing oracle, never outputs an invalid signature, and can forge every chosen message whenever it succeeds. Its success probability over the key and attack coins is at least

$$1 - \delta_{\text{key}} - \alpha_{V,s_{0.3}^{+}} - \beta_{V,s_{0.3}^{+}}, \tag{42}$$

where δ_{key} is the key-level exceptional probability in Corollary B.19. In particular, the success probabilities at Levels I, III, and V are respectively at least

$$0.999999996259, \qquad 0.999999999998, \qquad 0.999999964716.$$

Proof. Outside the key-level exceptional set, the projective discriminant and a deterministic pair-line witness both exist. A successful checked solver pass returns the complete core, so the planted point survives exact filtering and yields the planted expanded decomposition. The deterministic signing scan then returns a certificate that inverts every public target. Apply Corollary 3.6 and union-bound the key, outer-discriminant, and internal-solver failures. Exact verification ensures that failure affects only whether a forgery is returned, never whether a returned forgery is valid. □

C Concrete solver derivations and secondary ledgers

This appendix preserves the complete arithmetic derivations behind the concrete rows in the main paper: flattened dense lifting, explicit sparse working fields, characteristic-polynomial degree control without global separation, local target separation, exact Frobenius descent, root success, and the full Level-I component and sensitivity ledgers.

C.1 Dense-quadratic batching and flattened lifting

C.1.1 Why the direct straight-line-program bound is not the last word

For dense quadrics, a direct straight-line program has size $L = O(V^3)$. Substituting it into (48) gives

$$\widetilde{O}(V^5 4^V) \tag{43}$$

operations in K'. This is a valid generic specialization, but it repeatedly charges contractions that all use the same dense quadratic coefficient tensor.

Write

$$f_i(X) = X^{\mathsf{T}} A_i X + b_i^{\mathsf{T}} X + c_i. \tag{44}$$

At precision k, the lifted coordinates live in a coefficient algebra

$$\mathcal{A}_k := K'[T,U]/(T^k, Q_k(T,U)), \tag{45}$$

where Q_k is monic of U-degree at most C. The V coordinate vectors form a $V \times (Ck)$ coefficient matrix. Flattening the matrices A_i turns all Jacobian contractions into one rectangular matrix product over K'. Quadratic values follow from the Jacobian and X, while the Newton solve uses matrix algebra over $\mathcal{A}_k$.

Proposition C.1 (Dense-quadratic lifting ledger)**.** Let ω be an admissible square-matrix multiplication exponent, and let $\mu(C, k)$ be the cost of one multiplication in $\mathcal{A}_k$. The precision-k work of the global Newton lift can be organized using

$$\widetilde{O}(V^3 Ck + (V^\omega + V^2)\mu(C, k)) \tag{46}$$

operations in K'. Summed over precision doubling through $2C_0$, this gives

$$\widetilde{O}(V^3 CC_0 + (V^\omega + V^2)\mu(C, 2C_0)). \tag{47}$$

Proof. Stack the V matrices A_i into a $V^2 \times V$ scalar matrix. Multiplication by the $V \times (Ck)$ lifted-coordinate matrix forms every linear Jacobian contraction simultaneously in $O(V^3 Ck)$ scalar operations. This term is written explicitly rather than hidden inside the coefficient-algebra multiplication count.

The identity

$$f_i(X) = \frac{1}{2} X^\mathsf{T} (2A_i X + b_i) + \frac{1}{2} b_i^\mathsf{T} X + c_i$$

computes all values with $O(V^2)$ products in $\mathcal{A}_k$. Newton matrix inversion or solving costs $O(V^\omega)$ such products, and primitive-element renormalization lies within the same envelope. The terminal precision is $2C_0$; quasi-linear multiplication makes the last doubling stage dominate up to logarithmic factors. □

C.1.2 One multiplication, not a product of two multiplication costs

The earlier stress analysis bounded multiplication in (45) by applying a quasi-linear polynomial-multiplication bound first in U and then in T. This gives a product $\mathsf{M}(C)\mathsf{M}(k)$. A bivariate array can instead be flattened once.

Lemma C.2 (Collision-free Kronecker flattening)**.** Let $R_k = K'[T]/(T^k)$ and let $Q_k(U) \in R_k[U]$ be monic of degree C. After precomputing the reciprocal of the reversed Q_k, one multiplication in $R_k[U]/(Q_k)$ can be performed using at most three ordinary polynomial multiplications whose input length is below $2Ck$, plus $O(Ck)$ additions. The reciprocal precomputation costs $O(\mathsf{M}(2Ck) \log C)$ operations in K'.

Proof. For a polynomial

$$A(U, T) = \sum_{0 \le i < C,\ 0 \le j < k} a_{i,j} U^i T^j,$$

choose the radix $B = 2C - 1$ and encode

$$U^i T^j \longmapsto Z^{i+Bj}. \tag{48}$$

In a product, the U-exponent is at most $2C - 2 = B - 1$. It therefore never carries into the T digit. The largest input exponent is $(C - 1) + B(k - 1) < Bk < 2Ck$, so one ordinary univariate multiplication recovers the complete bivariate product modulo T^k.

To reduce the U-degree modulo the monic polynomial Q_k, use the standard reciprocal division algorithm over the coefficient ring R_k. With the reciprocal precomputed, one product obtains the quotient estimate and one product forms the quotient times Q_k. Together with the unreduced product, this is three products of the same padded length. Newton inversion of the reversed monic polynomial gives the stated one-time reciprocal cost over any commutative coefficient ring. □

Corollary C.3 (Flattened lifting complexity)**.** Let $\mathsf{M}(n)$ denote a quasi-linear polynomial-multiplication cost. The batched lift can be implemented in

$$\widetilde{O}(V^3 CC_0 + (V^\omega + V^2)\mathsf{M}(CC_0)) = \widetilde{O}(V^{\omega+1}4^V) \tag{49}$$

operations in K'.

For the explicit stress function

$$\mathsf{M}(n) := n \log_2 n \log_2 \log_2 n, \tag{50}$$

the entire precision-doubling schedule for one coefficient-algebra product is bounded by

$$3\mathsf{M}(8CC_0). \tag{51}$$

Proof. Apply Lemma C.2 at precisions $2C_0, C_0, C_0/2, \ldots$. At the terminal stage, each encoded input has length below $4CC_0$. Monotonicity of the logarithmic factors gives

$$\sum_{j\geq 0} \mathsf{M}\left(\frac{4CC_0}{2^j}\right) < 2\mathsf{M}(4CC_0) < \mathsf{M}(8CC_0).$$

There are three ordinary products per modular multiplication. Reciprocal precomputation contributes an additional logarithmic factor but is dominated by the explicit V^2 outer allowance. Finally, $CC_0 = 2^{2V-1}(V+2)$. □

Remark C.4 (What was improved and what was corrected). Flattening does not alter the exponential factor 4^V; it replaces the product of two independent quasi-linear overheads by the overhead at the total coefficient-array length. The factor 8 in (51) is important: an earlier draft stopped at precision C_0, whereas rational reconstruction requires $2C_0$. All finite values in Section 10 use the corrected precision.

C.2 Sparse working-field models

Both solvers may choose any convenient representation of their finite working field. The directional solver starts over $K = \mathbb{F}_{127^3}$, so its auxiliary fields must contain K; the direct solver starts over $\mathbb{F}_{127}$ and has no such divisibility requirement. Projectivization lowers the Level-I direct working degree from 16 to 15.

Lemma C.5 (Explicit sparse auxiliary fields)**.** The following polynomials are irreducible over $\mathbb{F}_{127}$:

$$\phi_{15}(X) = X^{15} + X^{12} - X + 1, \qquad \phi_{18}(X) = X^{18} + X^4 + 1,$$
$$\phi_{22}(X) = X^{22} + X + 1, \qquad \phi_{24}(X) = X^{24} + X^5 - X + 1,$$
$$\phi_{30}(X) = X^{30} + X^7 + 1.$$

The projective direct-forgery solver uses degrees $15, 22, 30$ at Levels I, III, and V. The directional solver uses degrees $18, 24, 30$.

Proof. The companion audit applies Rabin's irreducibility criterion [26]. For every displayed degree d, it verifies

$$X^{127^d} = X \pmod{\phi_d}$$

and, for each prime divisor e of d,

$$\gcd(\phi_d, X^{127^{d/e}} - X) = 1.$$

All computations are exact over $\mathbb{F}_{127}$, and the machine-readable output records every gcd degree. The same audit certifies the additional sparse fields used in the retuning table of Table 4, including $X^{26} + X^3 - X + 1$. □

Schoolbook convolution is unnecessary at these modest extension degrees. The next statement freezes an explicit recursive circuit rather than invoking an asymptotic multiplication theorem.

Proposition C.6 (Audited Karatsuba extension-multiplication envelope)**.** Let

$$g_{\mathrm{mul}} = \mathsf{G}_{\mathrm{mul}}(\log_2 127), \qquad g_{\mathrm{add}} = 4\log_2 127.$$

For length-d coefficient vectors, let (M_d, A_d) be chosen recursively as follows. The schoolbook candidate is

$$(d^2, (d-1)^2).$$

For $d \geq 2$, put $a = \lceil d/2 \rceil$ and $b = \lfloor d/2 \rfloor$. The Karatsuba candidate is

$$\big(2M_a + M_b,\ 2A_a + A_b + 4d - 4\big). \tag{52}$$

Choose whichever candidate has smaller value $Mg_{\mathrm{mul}} + Ag_{\mathrm{add}}$. If s_d is the number of nonleading terms of ϕ_d, one multiplication in $\mathbb{F}_{127^d}$ followed by sparse reduction is then charged by

$$\mathsf{G}^{\mathrm{sparse}}_{127^d} := M_d g_{\mathrm{mul}} + \big(A_d + s_d(d-1)\big)g_{\mathrm{add}}. \tag{53}$$

For the official working fields, the selected plans are:

d	M_d	A_d	s_d
15	81	332	3
18	153	404	2
22	225	480	2
24	243	512	3
30	243	1112	2

Proof. The schoolbook counts are exact: d^2 coefficient products and $d^2 - (2d-1) = (d-1)^2$ additions. For the recursive candidate, write each operand as a low block of length a plus X^a times a high block of length b. The products of the two low blocks and of their sums have length a; the high-block product has length b, giving $2M_a + M_b$ coefficient multiplications. Forming the sums, subtracting the low and high products from the middle product, and combining the three shifted outputs uses at most $4d-4$ additional coefficient additions, giving (52). Recursion therefore realizes the displayed pair (M_d, A_d).

The unreduced product has only $d-1$ high coefficients. Eliminating each one modulo ϕ_d uses s_d additions or subtractions and no nontrivial constant multiplication because every nonleading coefficient is $+1$ or -1. This proves (53). The companion audit executes the same recursive plan and compares 48 random products against schoolbook convolution before certifying the gate counts. □

C.3 Characteristic polynomials without global separation

Let $\Gamma \subset \mathbb{A}^{1+a}$ be the union of the saturated homotopy-curve components that dominate the t-line, with degree at most $D = C_0$. The Jacobian saturation removes the non-etale locus, so over $K'(t)$ its generic fiber is a finite product of separable field extensions,

$$B = K'(t)[X_1, \ldots, X_a]/J, \qquad \dim_{K'(t)} B = C.$$

Write x_j for the image of X_j and M_{x_j} for multiplication by x_j on B.

Introduce independent coefficients $\Lambda = (\Lambda_1, \ldots, \Lambda_a)$ and the generic linear element

$$u_\Lambda = \sum_j \Lambda_j x_j.$$

Its characteristic polynomial is

$$\chi_\Lambda(t, U) = \det\left(UI - \sum_j \Lambda_j M_{x_j}\right). \tag{54}$$

Schost proves two facts that are exactly suited to the present use: specialization of Λ gives the characteristic polynomial of the specialized linear element, and the generic characteristic polynomial can be written

$$\chi_\Lambda = \frac{\Xi_\Lambda}{\Theta},$$

where the t-degrees of Ξ_Λ and Θ are at most $\deg \Gamma \le D$ and Θ is independent of Λ [28, Lemmas 2–3 and Proposition 2]. This is the characteristic-polynomial, or generic u-resultant, form of the parametric degree theorem.

Schost phrases one corollary using a primitive element with coefficients in the ground field, together with a field-size hypothesis. The generic characteristic polynomial itself does not require such an element. After extending to an algebraic closure of $K'(t)$, the finite etale algebra has C distinct geometric points. For every distinct pair, the difference

$$\Lambda \cdot (x_s - x_r)$$

is a nonzero polynomial in the independent indeterminates Λ. Hence u_Λ separates the geometric points over the rational-function field $K'(t)(\Lambda)$ and is primitive there, irrespective of the size of K'. Apply Schost's degree argument after this transcendental scalar extension. The determinant in (54) is formed from the original multiplication matrices, so its coefficients already lie in $K'(t)[\Lambda, U]$; scalar extension cannot lower a reduced t-degree and thereby hide a larger degree over K'. Thus no K'-rational global separator, and in particular no $|K'| \gg C^2$ assumption, is used.

Lemma C.7 (Nonseparating eliminant)**.** For every $\lambda \in K'^a$, including a colliding one,

$$q_\lambda(t, U) := \prod_{s=1}^{C} (U - \lambda(x_s(t))) = \chi_\Lambda(t, U)\big|_{\Lambda=\lambda}.$$

Every coefficient of q_λ has a common-denominator representation with numerator and denominator t-degrees at most D.

Proof. The first identity is the characteristic-polynomial product formula in the reduced generic fiber, with repeated factors allowed after specialization. The second assertion follows by substituting $\Lambda = \lambda$ in Ξ_Λ/Θ. Since Θ is independent of Λ, a separator collision cannot create a new parameter denominator or increase its degree. □

The coordinate interpolation numerators are already encoded in the same generic characteristic polynomial.

Lemma C.8 (Characteristic-polynomial polar)**.** Define

$$v_j(t, U) = \sum_s x_{s,j}(t) \prod_{r \neq s} (U - \lambda(x_r(t))).$$

Then

$$v_j(t, U) = -\left.\frac{\partial \chi_\Lambda(t, U)}{\partial \Lambda_j}\right|_{\Lambda=\lambda}. \tag{55}$$

Consequently each v_j, and every fixed affine-linear combination of the v_j and q_U, has numerator and denominator t-degrees at most D, without assuming that λ separates the branches.

Proof. Over a splitting field, differentiate

$$\chi_\Lambda(t,U) = \prod_s \left(U - \sum_i \Lambda_i x_{s,i}(t) \right)$$

with respect to Λ_j and then specialize $\Lambda = \lambda$. This gives (55). Differentiating Ξ_Λ in the linear-form coefficients does not increase its t-degree, while the denominator Θ is unchanged. $\square$

This directly covers the sole linear sketch used later:

$$b_1(P) = \lambda^{q^{d-1}} \cdot P.$$

Its interpolation numerator is the corresponding fixed K'-linear combination of the v_j.

For the rational oil filter

$$\psi(X) = \frac{H_0(X)}{\delta + \lambda(X)},$$

the preceding split-before-lift note proves that the saturated graph curve has degree at most $2D$. Apply the same construction to its generic linear element

$$u_{\Lambda,\mu} = \Lambda \cdot X + \mu Y,$$

where $Y = \psi(X)$ is the graph coordinate. Specializing $(\Lambda, \mu) = (\lambda, 0)$ gives q_λ, and

$$-\left.\frac{\partial \chi_{\Lambda,\mu}}{\partial \mu}\right|_{(\lambda,0)} = v_\psi.$$

The graph degree bound therefore gives numerator and denominator t-degrees at most $2D$ for the filter numerator. Consequently the existing precision

$$P = 4C_0 + 1$$

remains valid.

Remark C.9. Separation is needed only when one wants the linear element to be a primitive coordinate for every point. The attack instead keeps the full characteristic polynomial and later accepts only a simple target factor. Characteristic polynomials and their coefficient polars remain defined at colliding linear forms.

C.4 Local separation at the target point

At target specialization, normalize q and each scalar numerator by the common t-valuation exactly as in Safey El Din–Schost Lemma 12 [27]. Call λ *valuation-generic* if every branch escaping to infinity with minimum coordinate valuation $\mu_s < 0$ satisfies

$$\operatorname{ord} \lambda(x_s) = \mu_s.$$

This condition, not pairwise branch separation, is what the specialization proof uses to remove the infinite factors from the specialized eliminant.

Lemma C.10 (Simple-factor recovery)**.** Assume λ is valuation-generic. Let $\bar{q}(U)$ and $\bar{v}_b(U)$ be the normalized target specializations. If $\tau \in K'$ is a simple root of $\bar{q}$, then there is exactly one finite represented target point P with $\lambda(P) = \tau$, and

$$b(P) = \frac{\bar{v}_b(\tau)}{\bar{q}'(\tau)}.$$

No separation condition is needed away from that one fiber value.

Proof. Let S_∞ and S_0 index the escaping and finite branches. With

$$e = -\sum_{s \in S_\infty} \mu_s,$$

valuation genericity gives

$$(t^e q)(0, U) = \gamma \prod_{s \in S_0} (U - \lambda(P_s))$$

for a nonzero scalar γ. In the normalized numerator, the summand indexed by a finite branch s specializes to

$$\gamma b(P_s) \prod_{r \in S_0,\ r \neq s} (U - \lambda(P_r)).$$

A summand indexed by an escaping branch need not vanish term by term; rather, every such surviving summand is divisible by the full finite product $\prod_{r \in S_0}(U - \lambda(P_r))$. It therefore vanishes modulo $\bar{q}$, and in particular at every finite root of $\bar{q}$. Thus $\bar{v}_b$ modulo $\bar{q}$ is the ordinary interpolation numerator on the finite limits. A simple root τ belongs to exactly one finite branch, and evaluation followed by division by $\bar{q}'(\tau)$ gives the claimed value. $\square$

Condition on the existence of a canonical usable nonsingular base-field target point $P_\star$; for example, choose the first under any fixed ordering. Nonsingularity makes it an isolated multiplicity-one point. Let I generic branches escape to infinity and let F have finite target limits, counted with branch multiplicity. Then $I + F = C$. There are at most I valuation-cancellation hyperplanes and at most $F - 1$ collision hyperplanes against $P_\star$. Each bad equality

$$\lambda(w) = 0 \quad \text{or} \quad \lambda(P - P_\star) = 0$$

cuts out at most $|K'|^{a-1}$ choices of λ in $(K')^a$. Thus a uniform $\lambda \in (K')^a$ fails valuation genericity or collides at $P_\star$ with probability at most

$$\frac{I + F - 1}{|K'|} = \frac{C - 1}{|K'|}. \tag{56}$$

The coefficient vector of the equation may lie only over an algebraic closure, so its solution set need not literally be a K'-defined hyperplane. The count is still valid: choose a nonzero coordinate of that vector and fix all other coefficients of λ; the remaining equation has at most one solution in K' (and may have none).

For the filter denominator, first compute the C known start values $\lambda(\xi_s)$ and sample δ uniformly from the complement of the forbidden set $\{-\lambda(\xi_s)\}_s$ in K'. Rejection sampling costs an expected factor at most $(1 - C/|K'|)^{-1}$ for a scan of the start values; this scan is negligible beside one length-P filter lift and is included in the miscellaneous envelope. Conditional on this choice, only the at most $F \le C$ finite target values remain forbidden. Therefore

$$\Pr[\delta + \lambda(P) = 0 \text{ for a finite target branch}] \le \frac{C}{|K'| - C}. \tag{57}$$

The old $C^2/|K'|$ event has disappeared.

C.5 Exact one-sketch Frobenius descent

Set

$$b_1(P) = \lambda^{q^{d-1}} \cdot P, \qquad K' = \mathbb{F}_{q^d}.$$

The separator value $\tau = \lambda(P)$ is already available from the factor of $\bar{q}$, so no numerator for $\lambda \cdot P$ is needed.

Lemma C.11 (Simple-root descent is exact)**.** Let $\tau \in K'$ be a simple linear factor of the target eliminant, and let P be its unique represented point. Then

$$b_1(P)^q = \tau \quad \Longleftrightarrow \quad P \in \mathbb{F}_q^a.$$

Proof. The forward identity for a base-field point is immediate. Conversely,

$$b_1(P)^q = \lambda \cdot P^q.$$

If it equals $\tau = \lambda \cdot P$, then P and P^q have the same separator value. The target equations and homotopy are defined over $\mathbb{F}_q$, so the represented finite target limits are stable under Frobenius. If $P^q \neq P$, the two distinct represented points produce multiplicity at least two at $U = \tau$, contradicting simplicity. Hence $P^q = P$. □

This eliminates the probabilistic $C/|K'|$ false-positive term. It also eliminates the abort cap: every simple factor passing the Frobenius test and the nonzero-square oil filter is a genuine base-field target solution. The algorithm reconstructs coordinates for the first such factor and stops, so exactly one coordinate re-lift is charged per successful forgery.

C.6 Root success probability

For completeness, this pass expands the probability calculation inherited from the split-before-lift note. Let M count all off-oil affine roots of the normalized residual system, and let N count those with invertible affine Jacobian. The parent manuscript proves, for an actual parameter $\mu \geq 1 - q^{-1}$,

$$\mathbb{E}[M] = \mu, \qquad \mathbb{E}[M^2] = 2\mu + \mu^2 - (q-1)\mu q^{-r}, \qquad \mathbb{E}[N] = \mu\pi,$$

where $\pi = \prod_{j=1}^{r-1}(1 - q^{-j})$. The roots occur in pairs $z, -z$, and each nonsingular normalized pair yields one eligible projective point. Applying the second Bonferroni inequality to the projective-point events and bounding pairs of nonsingular roots by pairs of all roots gives

$$\begin{aligned} \Pr[\text{usable nonsingular projective root}] &\geq \frac{\mathbb{E}[N]}{2} - \mathbb{E}\left[\binom{M/2}{2}\right] \\ &= \frac{\mu\pi}{2} - \frac{\mu^2}{8} + \frac{(q-1)\mu q^{-r}}{8}. \end{aligned}$$

The right-hand side is increasing for $\mu \in [1 - q^{-1}, 1]$, so substituting $\mu = 1 - q^{-1}$ yields

$$p_{\text{root}} = 0.3690869822\ldots.$$

As a separate fallback, the ledger also substitutes the parent manuscript's already-proved lower bound $0.326336998767509\ldots$; the Level-I total then becomes 140.531461, still 2.469 bits below target.

C.7 Finite Level-I ledger

At Level I the local variant has two scalar families, b_1 and ψ, instead of three. The terminal product tree therefore uses

$$\frac{a}{2}(1 + 2 \cdot 2) = 125$$

long-product units. Rational reconstruction contributes 99.885 units. Candidate coordinate evaluation is charged once over the complete Las Vegas execution, rather than per trial, and

Table 1: Level-I component ledger for extension degree $d = 8$.

Component	$\log_2$ gates
Initial base-field branch lifts	139.358
Valuation-safe filter series	132.120
Terminal scalar recombination	138.791
One on-demand coordinate re-lift	137.860
Extension-valued leaf formation	127.610
Coordinate-rerun separator leaves	124.898
Factorization and exact verification envelope	115.707
Preprocessing	63.365
Total	**140.383995**

the 64-unit miscellaneous cushion is retained. This pass also charges the one-time formation of $u_s = \lambda \cdot x_s$ during that coordinate rerun.

The one-trial success lower bound is 0.353997, giving restart factor 2.824886. At $d = 8$, the local-separator failure bound in (56) is 0.016637, and the conditional target-denominator bound in (57) is 0.016918.

Table 2: Level-I sensitivity to the multiplier on $\mathsf{M}(n) \log n$ in rational reconstruction, retaining $d = 8$.

Multiplier	8	10	16	32	64	128
Total	140.384	140.424	140.538	140.805	141.225	141.815

Table 3: Level-I cross-checks. Each row reoptimizes the extension degree.

Online-envelope multiplier	1	2	4	8	16
Best extension degree	8	8	9	9	9
Total	140.384	141.117	141.952	142.856	143.806

Thus the candidate remains below 2^{143} throughout the rational-reconstruction sensitivity range. It also remains below target after multiplying the entire claimed online-product envelope by eight, but not by sixteen; the finite margin therefore does not excuse the remaining obligation to provide a fully explicit schedule. The aggregate Level-I margin corresponds to an overall multiplicative cushion of $2^{2.616} = 6.13$ in the inherited gate model; this does not address memory traffic or a change of model.

D Deterministic fallbacks for equivalent-key recovery

The main paper proves that one directional root with full-rank derivative determines the hidden graph and states the rank probability used by the verified attack. This section gives the exact public-random calculations, the fallback that combines roots with partial derivative rank, and a deterministic list of square subsystems.

D.1 Exact direction-rank calculations

For a fixed nonzero $c \in K^O$, Definition 3.1 makes each vector $B_i c$ uniform in K^V, independently across i. Thus $\mathcal{B}(c)$ is a uniform $m \times V$ matrix and has full column rank with probability $\rho_{m,V}(Q)$ from (40). A preselected coordinate direction therefore fails with probability $\tau_{m,V}(Q)$, whose base-two logarithm is

$$-62.898161, \qquad -62.898161, \qquad -83.864216$$

at Levels I, III, and V.

For the coordinate directions, the vectors $B_i e_j$ use separate columns of the independent matrices B_i. Hence $\mathcal{B}(e_1), \dots, \mathcal{B}(e_O)$ are independent. This proves (42); its base-two logarithms are

$$-1132.166907, \qquad -1635.352198, \qquad -2935.247544.$$

D.2 Combining roots with partial derivative rank

Proposition D.1 (Several directions can be combined)**.** Let $c_1, \dots, c_r \in K^O$ and suppose that roots $x_j = Gc_j$ of the corresponding public direction systems have been found. Stack their public completion matrices and right-hand sides:

$$\mathcal{D}_{\boldsymbol{c}}(\boldsymbol{x}) := \begin{pmatrix} \mathcal{D}_{c_1}(x_1) \\ \vdots \\ \mathcal{D}_{c_r}(x_r) \end{pmatrix}, \qquad b_{\boldsymbol{c},d}(\boldsymbol{x}) := \begin{pmatrix} b_{c_1,d}(x_1) \\ \vdots \\ b_{c_r,d}(x_r) \end{pmatrix}.$$

If the stacked matrix has column rank V, then for every $d \in K^O$ the public system

$$\mathcal{D}_{\boldsymbol{c}}(\boldsymbol{x})y = b_{\boldsymbol{c},d}(\boldsymbol{x})$$

has the unique solution $y = Gd$. Thus complementary derivative information from several recovered roots determines the graph even when no individual direction has full rank.

Proof. For each j, the proof of Proposition 5.2 shows that Gd satisfies the jth block. It therefore satisfies the stacked system. Full column rank gives uniqueness. □

D.3 A deterministic list of square systems

The directional system has m equations in V variables. Trying every row subset gives $\binom{m}{V}$ square systems. The following construction replaces that list with $V(m-V)+1$ public linear recombinations.

Index the equations by $r = 0, \dots, m-1$. For $0 \le a < V$, write $(r)_a = r(r-1)\cdots(r-a+1)$, with $(r)_0 = 1$, and define

$$\mathsf{C}(t)_{a+1,r+1} := (r)_a t^{r-a} \qquad (t \in K), \tag{58}$$

where the entry is zero when $r < a$. The first V columns are triangular with diagonal $0!, 1!, \dots, (V-1)!$. They are nonsingular because $V - 1 < 127 = \operatorname{char} K$.

Lemma D.2 (A one-parameter rank condenser)**.** Let $B \in K^{m \times V}$ have column rank V, where $m \le 127$. Then

$$W_B(T) := \det(\mathsf{C}(T)B) \tag{59}$$

is a nonzero polynomial of degree at most $V(m-V)$.

Proof. For column j of B, form $f_j(T) = \sum_{r=0}^{m-1} B_{r+1,j} T^r$. The columns of B are independent exactly when $f_1, \dots, f_V$ are independent, and $\mathsf{C}(T)B$ is their ordinary Wronskian matrix.

Choose a basis $g_1, \dots, g_V$ of their span with distinct increasing degrees $0 \le d_1 < \cdots < d_V < m$. Up to the nonzero product of their leading coefficients, the leading coefficient of their Wronskian is

$$\det\big((d_j)_a\big)_{0 \le a < V,\, 1 \le j \le V} = \prod_{i<j} (d_j - d_i).$$

This value is nonzero because the d_j are distinct and below 127. A change of basis only multiplies the Wronskian by a nonzero scalar. Finally,

$$\deg W_B \le \sum_{j=1}^{V} d_j - \frac{V(V-1)}{2} \le V(m-V),$$

where the last bound uses the V largest distinct degrees below m. □

Conditional on $\operatorname{rank} \mathcal{B}(c) = V$, a uniform $t \leftarrow K$ therefore satisfies

$$\Pr[\mathsf{C}(t)\mathcal{B}(c) \text{ is invertible}] \ge 1 - \frac{V(m-V)}{Q}. \tag{60}$$

Corollary D.3 (A short deterministic public list)**.** Fix any public list of $V(m-V)+1$ distinct elements of K. Whenever $\mathcal{B}(c)$ has column rank V, at least one t on the list makes the square system

$$\mathsf{C}(t)\,(q_{1,c}, \dots, q_{m,c})^{\mathsf{T}} = 0 \tag{61}$$

have the planted root as a nonsingular solution.

For one coordinate direction, the list sizes at Levels I, III, and V are

$$105, \qquad 153, \qquad 307.$$

Scanning every coordinate direction uses

$$1{,}890, \qquad 3{,}978, \qquad 10{,}745$$

square systems and fails in the public-random model only with probability $\tau_{m,V}(Q)^O$. Thus all outer choices can be deterministic.

Proof. The polynomial in (59) has degree at most $V(m-V)$ and cannot vanish at every point of a longer list. At the planted root, the Jacobian of (61) is $-2\mathsf{C}(t)\mathcal{B}(c)$ by Proposition 5.1. The numerical counts use $m - V \in \{2, 2, 3\}$. □

Corollary D.4 (Category-scale deterministic direction lists)**.** Use the first

$$r_{\mathrm{dir}} = 3, \qquad 4, \qquad 4$$

coordinate directions at Levels I, III, and V, and use the condenser list for each direction. The resulting public lists contain

$$315, \qquad 612, \qquad 1228$$

square systems. In the public-random model, the probability that none has the planted point as a nonsingular root is $\tau_{m,V}(Q)^{r_{\mathrm{dir}}}$, whose base-two logarithms are

$$-188.694, \qquad -251.593, \qquad -335.457. \tag{62}$$

These bounds are below the three category exponents $143, 207, 272$.

Proof. The coordinate-direction matrices are independent by Proposition 5.4. If one of the selected matrices has full column rank, Corollary D.3 supplies a value with nonsingular planted Jacobian. Multiplying the one-direction failure logarithm by r_{dir} gives (62); multiplying the list lengths gives the displayed system counts. $\square$

Remark D.5 (The list length is geometrically optimal). The Grassmannian of V-subspaces of an m-dimensional space has dimension $V(m-V)$. For a fixed full-rank map $R : K^m \to K^V$, the subspaces on which R is singular form a hyperplane section in Plücker coordinates. Over an algebraic closure, at most $V(m-V)$ such sections always have a common point. Hence no family of at most $V(m-V)$ recombinations works for every full-rank $m \times V$ matrix after all field extensions. The list above meets this lower bound exactly.

Sampling one field element t is convenient for the expected-cost theorem. The deterministic list removes that choice and is much smaller than the list of all V-row subsets.

E Direct-forgery proofs and preprocessing

This section gives the local distribution proofs, dimension counts, and finite preprocessing ledger used by the direct attack in Section 4. The main paper retains the complete reduction and success statements. We use the notation of Lemma 4.6 and (26): $E = K^{V+O}$, $\widehat{\mathcal{O}} \subset E$ is the hidden base-field oil space, and $\gamma \in K^*$ is fixed.

E.1 Proof of local value and derivative randomness

We prove Lemma 4.6. Write an off-oil point as $x = (v_x, o_x)$ in central K-coordinates, so $v_x \neq 0$. Choose a K-linear functional $\varphi : E \to K$ with $\varphi|_{\widehat{\mathcal{O}}} = 0$ and $\varphi(x) = 1$. The map

$$h \longmapsto (u \mapsto \mathrm{Tr}_{K/\mathbb{F}_q}(\gamma h(u)))$$

from K-linear maps $E \to K$ to $\mathbb{F}_q$-linear maps $E \to \mathbb{F}_q$ is an isomorphism by nondegeneracy of the trace pairing and equality of dimensions.

Fix a desired pair (c, λ) with $\lambda(x) = 2c$. Choose a K-linear h whose trace functional restricts to $\lambda/2$ on U, and define

$$b_h(u, w) = \varphi(u)h(w) + h(u)\varphi(w) - h(x)\varphi(u)\varphi(w).$$

This symmetric K-bilinear form vanishes on $\widehat{\mathcal{O}} \times \widehat{\mathcal{O}}$ and satisfies $b_h(x, u) = h(u)$. Hence

$$p_h(u) := b_h(u, u), \qquad Q_h(u) := \mathrm{Tr}_{K/\mathbb{F}_q}(\gamma p_h(u))$$

is a pulled-back UOV form with

$$Q_h(x) = \mathrm{Tr}(\gamma h(x)) = c, \qquad d(Q_h)_x(u) = 2\,\mathrm{Tr}(\gamma h(u)) = \lambda(u).$$

This proves the value and derivative claims.

For the last claim, suppose evaluation at off-oil points x and z is proportional by $\rho \in \mathbb{F}_q^*$. Comparing central coefficients and using the trace pairing gives

$$v_z v_z^\top = \rho v_x v_x^\top, \qquad v_z o_z^\top = \rho v_x o_x^\top.$$

The first identity gives $v_z = \lambda v_x$ and $\lambda^2 = \rho$ for some $\lambda \in K^*$. The second then gives $o_z = \lambda o_x$. Finally,

$$\gcd(2(q-1), q^3 - 1) = q - 1$$

because $q^2 + q + 1$ is odd. Thus $\lambda^{q-1} = 1$ and $\lambda \in \mathbb{F}_q$.

E.2 The seed count and constructive extension

For completeness, the fixed-oil-space estimate in Proposition 4.2 is a direct projective count. There are $(q^m - 1)/(q - 1)$ projective oil points. A uniform seven-dimensional subspace contains any fixed projective point with probability $(q^7 - 1)/(q^n - 1)$. A union bound gives (17); substituting the three official pairs (n, m) gives (18).

We next give the full construction from Theorem 4.3. Suppose a j-dimensional common totally isotropic subspace W containing the seed point has already been constructed. Put

$$L_W = \bigcap_{i=1}^{s} W^{\perp_{B_{Q_i}}}.$$

The sj orthogonality conditions give

$$\dim(L_W/W) \geq n - (s+1)j.$$

For $j \leq r-1$, condition (19) makes this dimension at least $2s+1$. Choose any $(2s+1)$-dimensional subspace of the quotient. The s induced quadrics have total degree $2s < 2s+1$. Chevalley–Warning says that their number of common zeros is divisible by the characteristic [33, 34]. Since the zero vector is one zero, there is a nonzero zero. Exhaustive search over (20) finds one.

Lift this zero to $z \in L_W$. Then $Q_i(z) = 0$ and $B_{Q_i}(z, W) = 0$, so $W \oplus \langle z \rangle$ is again common totally isotropic. Iteration reaches dimension r.

For the official parameters, one seed search and the $r-1$ extension steps each evaluate three seven-variable quadrics on at most $|\mathbb{P}^6(\mathbb{F}_{127})|$ points. The companion audit also charges uniform seed generation, recomputation of every polar-orthogonal space by dense elimination, the random chart, and success normalization. The resulting preprocessing bounds at Levels I, III, and V are

$$2^{63.545}, \qquad 2^{64.077}, \qquad 2^{64.501}$$

Boolean gates.

E.3 Projectivization and Jacobian rank

We prove the claims in Corollary 4.5. For a nonzero common zero z of $H_1, \ldots, H_{r-1}$, replacing z by λz multiplies every quadratic value by λ^2. Some representative satisfies $H_0(\lambda z) = 1$ exactly when $H_0(z)$ is a nonzero square. In that case there are exactly two choices, λ and $-\lambda$.

Euler's identity gives $dH_i(z)(z) = 2H_i(z) = 0$ for $1 \leq i < r$, whereas $dH_0(z)(z) = 2$. The first $r-1$ Jacobian rows have rank $r-1$ exactly when their common kernel is the radial line $\langle z \rangle$. In that case the last row is not in their span, so the affine Jacobian has rank r. Conversely, an invertible affine Jacobian forces the first $r-1$ rows to have rank $r-1$. Finally, among the $q^r - 1$ nonzero linear forms on U, exactly $q^{r-1} - 1$ vanish at a fixed projective point. A chart containing the point preserves nonsingularity and gives the square affine system used by the solver.

E.4 Second-moment root calculation

We give the full calculation behind Proposition 4.7. The off-oil seed point belongs to U, so U is not contained in $\widehat{\mathcal{O}}$ and $d \leq r-1$. Let $S = U \setminus \widehat{\mathcal{O}}$, and let M count all roots of (27) in S. At any fixed $x \in S$, Lemma 4.6 gives

$$\Pr[x \text{ is a root}] = q^{-r}.$$

Hence $\mathbb{E}[M] = |S|q^{-r} = \mu$.

For two off-oil points, the evaluation functionals are independent unless the points are scalar multiples. If $z = \lambda x$ and both solve (27), homogeneity and the last equation give $\lambda^2 = 1$. The only dependent ordered pairs that can both solve the target are therefore $z = x$ and $z = -x$. The other $q - 3$ nontrivial base-field multiples have joint probability zero. Every nonproportional ordered pair has joint probability q^{-2r}. Consequently,

$$\mathbb{E}[M^2] = 2\mu + \mu^2 - (q-1)\mu q^{-r}.$$

Let N count only roots with invertible Jacobian. At a fixed root, the conditional derivative statement makes the Jacobian rows independent and uniform subject to Euler's relation. The target vector is nonzero, so one column is fixed and nonzero while the other $r - 1$ columns are uniform. Thus

$$\Pr[J(x) \text{ invertible} \mid x \text{ is a root}] = \pi_{r-1}(q), \qquad \mathbb{E}[N] = \mu\pi_{r-1}(q).$$

Since $N^2 \le M^2$, the second-moment inequality gives

$$\Pr[N > 0] \ge \frac{\mathbb{E}[N]^2}{\mathbb{E}[M^2]},$$

which is (29). The right-hand side increases with μ. Using $\mu \ge 1 - q^{-1}$ and substituting the three values of r gives (30).

E.5 Exact degree-three direct-forgery test

The companion program `QRUOV_direct_qr_toy.py` instantiates the degree-three model over $\mathbb{F}_{3^3}$ and performs the public attack over $\mathbb{F}_3$. After building the isotropic subspace, it enumerates the projective roots of the remaining zero-target forms and recovers the affine scale from the final equation. In the frozen 200-key run with

$$(V, O, n, m, r, k) = (3, 2, 15, 6, 3, 2),$$

199 keys yielded a verified forgery within twelve outer trials. Every returned vector passed both the base-field public map and an independent extension-field evaluator. Of the 200 final seed subspaces, 185 were disjoint from the hidden oil space, while every selected seed point was off oil. Seed-space disjointness is a sufficient production-scale guarantee, not a condition tested by the public algorithm. This experiment is a correctness test, not a production benchmark.

F Expanded-key certificate calculations

This section records the matrix and probability calculations behind Propositions 3.4 and 3.7. The main paper keeps the public certificate chain and the bounds used by later theorems. The derivations are collected here so that the attack flow does not stop for specification-level pull-back formulas or random-matrix counting.

F.1 Exact pull-back to the official verification map

Let $G' \in K^{V\times O}$ satisfy every graph identity (3). With the notation of Proposition 3.4,

$$S_{G'}^{-1} = \begin{pmatrix} I_V & -G' \\ 0 & I_O \end{pmatrix}.$$

A direct block multiplication gives

$$S_{G'}^{-\top} P_i S_{G'}^{-1} = \begin{pmatrix} A_i & E_i - A_iG' \\ E_i^\top - G'^\top A_i & C_i - G'^\top E_i - E_i^\top G' + G'^\top A_i G' \end{pmatrix}.$$

The lower-right block is zero by (3); the remaining blocks are exactly F_i'. Rearranging proves $P_i = S_{G'}^{\mathsf{T}} F_i' S_{G'}$.

We next connect this identity to the official base-field map. Let $\Phi_f : K \to \mathbb{F}_q^{\ell\times\ell}$ be the regular representation from the specification, let $W_f \in \mathbb{F}_q^{\ell\times\ell}$ be its invertible symmetrizer, and let ϕ be the blockwise inverse of Φ_f. Write $W_f^{(a)}$ for the block-diagonal matrix with a copies of W_f. If hats denote the corresponding base-field matrices, the forward pull-back of the specification has the form

$$P_i = \phi\Big((W_f^{(N)})^{-1}\widehat{P}_i\Big), \qquad F_i' = \phi\Big((W_f^{(N)})^{-1}\widehat{F}_i'\Big), \qquad S_{G'} = \phi(\widehat{S}_{G'}).$$

Hence the inverse pull-back sets

$$\widehat{P}_i = W_f^{(N)}\phi^{-1}(P_i), \qquad \widehat{F}_i' = W_f^{(N)}\phi^{-1}(F_i'), \qquad \widehat{S}_{G'} = \phi^{-1}(S_{G'}).$$

The pull-back identity converts the verified extension-field factorization into

$$\widehat{P}_i = \widehat{S}_{G'}^{\mathsf{T}} \widehat{F}_i' \widehat{S}_{G'}$$

for the unchanged base-field verification map [2, Sec. 4.1]. Signing uses this base-field pull-back, not the extension-field tuple with only $O = m/3$ oil variables.

F.2 Exact signing-search probabilities

Fix nonzero $\xi \in \mathbb{F}_q^{\ell}$. For each quotient-ring vinegar block j, let a_j place ξ in block j and zero elsewhere, and set $M_j = \mathsf{L}_{\widehat{F}}(a_j)$. Write $b_{i,j,k} \in K$ for entry (j,k) of the planted extension-field block B_i. The (i,k) block of the oil coefficient row produced by a_j is

$$2\xi^{\mathsf{T}} W_f \Phi_f(b_{i,j,k}) \in \mathbb{F}_q^{1\times\ell}.$$

The factor 2 is invertible. The $\mathbb{F}_q$-linear map

$$K \longrightarrow \mathbb{F}_q^{1\times\ell}, \qquad b \longmapsto \xi^{\mathsf{T}} W_f \Phi_f(b)$$

is injective. If $b \neq 0$, then $\Phi_f(b)$ is invertible and $\xi^{\mathsf{T}} W_f \neq 0$. Equal dimensions make the map an isomorphism. Each M_j is therefore uniform, and distinct j use disjoint entries of the independent blocks B_i. This proves the independence claim in Proposition 3.7.

For $0 \le r \le m$, the exact rank distribution of a uniform matrix is

$$\varrho_r := \Pr_{M \leftarrow \mathbb{F}_q^{m\times m}}[\operatorname{rank} M = r] = q^{-m^2} \prod_{i=0}^{r-1} \frac{(q^m - q^i)^2}{q^r - q^i}. \tag{63}$$

In particular,

$$\pi_m := \varrho_m = \prod_{k=1}^{m} (1 - q^{-k}), \qquad \sigma_m := 1 - \pi_m. \tag{64}$$

A basis test succeeds with probability π_m. Truncating the geometric random variable after V tests gives expected test count

$$\sum_{j=0}^{V-1} \sigma_m^j = \frac{1 - \sigma_m^V}{\pi_m} < \pi_m^{-1} < 1.008.$$

Independence gives failure probability σ_m^b for every fixed prefix of length b. Taking $b = V$ or the shorter values 22, 31, and 40 yields the basis-list probabilities stated in Proposition 3.7.

It remains to bound the deterministic pair-line fallback. Define

$$\chi_m := q^{-(m-1)} + (1 - q^{-(m-1)})q^{-1}, \qquad \zeta_{\text{pair}} := \varrho_{m-1}q^{-1}\chi_m + (1 - \varrho_m - \varrho_{m-1})(1 - \varrho_m). \quad (65)$$

Let A, B be independent uniform $m \times m$ matrices, and let $E_{A,B}$ be the event that their span contains no invertible matrix. If A is invertible, this event is impossible. If rank $A = m - 1$, invertible left and right changes of basis reduce A to $\operatorname{diag}(I_{m-1}, 0)$ without changing the distribution of B. Write

$$B = \begin{pmatrix} C & b \\ c^{\mathsf{T}} & d \end{pmatrix}.$$

On $E_{A,B}$, the polynomial $\det(A + tB)$ vanishes for every $t \in \mathbb{F}_q$. Its degree is at most $m < q$, so it is the zero polynomial. Its linear coefficient is d. After $d = 0$, its quadratic coefficient is $-c^{\mathsf{T}}b$. These conditions have probability at most $q^{-1}\chi_m$. If rank $A \le m - 2$, then $E_{A,B}$ implies that B is singular, an event of probability $1 - \varrho_m$. Conditioning on the rank of A gives

$$\Pr[E_{A,B}] \le \zeta_{\text{pair}}.$$

The chosen pairs are independent. Therefore

$$\eta_{\text{sign}} = \zeta_{\text{pair}}^{\lfloor V/2 \rfloor},$$

which gives the three numerical bounds in (15).

The basis matrices and $A + tB$ for $t \in \mathbb{F}_q^\times$ enumerate every projective point on each chosen pair line. The scan therefore succeeds whenever one pair span contains an invertible matrix and uses at most $V + \lfloor V/2 \rfloor (q - 1)$ tests. Finally, the determinant polynomial Δ in Proposition 3.7 has total degree m. If it is nonzero, Schwartz–Zippel gives success probability at least $1 - m/q$ for a uniform coefficient vector.

G Official seeded expansion: ideal primitives and fixed functions

The key-by-key theorem needs no expansion model. This appendix asks how often official seeded key generation satisfies the sufficient algebraic conditions for deterministic equivalent-key recovery. The official expansion uses public counter $2i - 1$ for the symmetric block A_i, public counter $2i$ for the rectangular block E_i, and secret counter 1 for the graph G [2, Secs. 4.7–4.10]. For every recommended set, the secret buffer length τ_2 is at most the public A_i buffer length τ_1. Hence, if the independently sampled public and secret seeds happen to coincide, the secret counter-1 raw stream is a prefix of the public A_1 stream. The resulting correlation between G and A_1 is expressly allowed by Definition 3.1; it is not a bad event.

For Levels I, III, and V, let $\lambda \in \{128, 192, 256\}$ be the seed length. The specification uses m public buffers of length τ_1 bytes and m public buffers of length τ_2 bytes. Each public rejection sampler exhausts its prescribed buffer with probability at most $2^{-\lambda}$. Put

$$\varepsilon_{\text{pub}} := 2m2^{-\lambda}. \quad (66)$$

No secret-buffer exhaustion term is needed: Definition 3.1 permits an arbitrary conditional law for G.

G.1 SHAKE in the ideal-XOF model

Proposition G.1 (Exact public-random transfer in the ideal-XOF model)**.** Model SHAKE as a random function from each domain-separated input to one consistent infinite output string. The total-variation distance between the official expanded-key distribution and the public-random model of Definition 3.1 is at most

$$\delta_{\mathrm{XOF}} := \varepsilon_{\mathrm{pub}} = 2m2^{-\lambda}.$$

This statement includes keys with equal public and secret seeds.

Proof. The $2m$ public seed/counter inputs are pairwise distinct, so their ideal-XOF strings are independent and uniform. If the seeds differ, the secret string is independent of all public strings. If they coincide, the secret counter-1 string is the required prefix of the public A_1 string; after rejection sampling, this induces an arbitrary conditional distribution of G given A_1, exactly as permitted by Definition 3.1. In either case, the even-counter E_i strings are independent uniform strings conditioned on (A, G). Abort only if one of the $2m$ public buffers contains too few accepted values. The union bound (66) pays for that event. Conditioned on no public exhaustion, the accepted A_i values have their product-uniform marginal and the accepted E_i values are product-uniform conditioned on (A, G), so the distribution is exactly Definition 3.1. □

G.2 AES in the ideal-cipher model

Let

$$T := m\left(\left\lceil \frac{\tau_1}{16} \right\rceil + \left\lceil \frac{\tau_2}{16} \right\rceil\right)$$

be the number of 128-bit public cipher blocks, and define

$$\delta_{\mathrm{perm}}(T) := 1 - \prod_{j=0}^{T-1}\left(1 - \frac{j}{2^{128}}\right), \qquad \delta_{\mathrm{IC}} := \delta_{\mathrm{perm}}(T) + \varepsilon_{\mathrm{pub}}. \tag{67}$$

Proposition G.2 (Exact public-random transfer in the ideal-cipher model)**.** Model AES as an ideal cipher with an independent random permutation for every key. The total-variation distance between the official expanded-key distribution and the public-random model of Definition 3.1 is at most δ_{IC}. This statement also includes equal public and secret AES keys.

Proof. For every fixed public key, the specification's T public counter blocks are distinct. Their ideal-cipher outputs are therefore a uniform ordered sample without replacement from 2^{128} blocks. The exact total-variation distance from T independent uniform blocks is the collision probability of the independent sample, namely $\delta_{\mathrm{perm}}(T)$.

Couple the without-replacement public block vector to an independent-block vector at this distance. If the public and secret keys differ, generate G from the independent secret-key permutation in both experiments. If the keys coincide, generate G from the prescribed prefix of the coupled public A_1 block vector in each experiment. In both cases this is a common randomized postprocessing kernel, so total variation cannot increase. Under the independent-block vector, A has the product-uniform marginal and E is product-uniform conditioned on (A, G). Truncation and rejection sampling are further postprocessing. Finally, public-buffer exhaustion contributes at most (66). □

G.3 Independent-graph transfer for direct forgery

The direct residual-root theorem uses Definition 3.2, so equal public and secret seeds or AES keys must now be paid for rather than absorbed as an allowed graph–coefficient correlation. Define

$$\delta_{\mathrm{XOF}}^{\mathrm{ind}} := \delta_{\mathrm{XOF}} + 2^{-\lambda}, \qquad \delta_{\mathrm{IC}}^{\mathrm{ind}} := \delta_{\mathrm{IC}} + 2^{-\lambda}. \tag{68}$$

Table 4: Official public-buffer parameters and exact ideal-primitive transfer bounds. The last column gives base-two logarithms. No seed-collision term is present.

Level	λ	m	τ_1	τ_2	T	$\log_2 \delta_{\mathrm{XOF}}$ / $\log_2 \delta_{\mathrm{IC}}$
I	128	54	4267	2916	24300	-121.245 / -99.863
III	192	78	9020	6123	73866	-184.715 / -96.655
V	256	105	16144	11018	178290	-248.286 / -94.112

Proposition G.3 (Independent-graph transfer in the ideal-primitive models)**.** The total-variation distance between official expanded-key generation and Definition 3.2 is at most $\delta^{\mathrm{ind}}_{\mathrm{XOF}}$ when SHAKE is modeled as an ideal XOF, and at most $\delta^{\mathrm{ind}}_{\mathrm{IC}}$ when AES is modeled as an ideal cipher.

Proof. Condition first on the independently sampled public and secret seeds (or AES keys) being distinct. This event fails with probability exactly $2^{-\lambda}$. On the conditioning event, the ideal primitive supplies an independent secret stream for G and independent public streams for A and E. The public buffer and, in the ideal-cipher model, without-replacement couplings are exactly those of Propositions G.1 and G.2. Their losses are δ_{XOF} and δ_{IC}, respectively. A union bound with the seed-collision event gives (68). ☐

For Levels I, III, and V, the corresponding base-two logarithms are

$$\log_2 \delta^{\mathrm{ind}}_{\mathrm{XOF}} = -121.232, -184.705, -248.279, \qquad \log_2 \delta^{\mathrm{ind}}_{\mathrm{IC}} = -99.863, -96.655, -94.112,$$

rounded upward in the last displayed digit.

Corollary G.4 (Official seeded direct-forgery invocation in ideal-primitive models)**.** Let p_{dir} denote the right-hand side of the direct-forgery success bound in (32). Under official seeded key generation, the success probability of the checked direct-forgery invocation is at least $p_{\mathrm{dir}} - \delta^{\mathrm{ind}}_{\mathrm{XOF}}$ in the ideal-XOF model and at least $p_{\mathrm{dir}} - \delta^{\mathrm{ind}}_{\mathrm{IC}}$ in the ideal-cipher model. Every returned signature remains valid against the unchanged verifier.

Proof. Total variation changes the probability of the ideal-model success event by at most the corresponding distance in Proposition G.3. Public verification is distribution-free and rejects every invalid candidate. ☐

G.4 From key-distribution distance to verified recovery

Let

$$\eta_{\mathrm{fast}} := \eta_{\mathrm{off}} + \eta_{\mathrm{oil}} + \eta_{\mathrm{span}} + \eta_{\mathrm{sign}}, \qquad \eta_{\mathrm{local}} := \tau_{m,V}(Q)^{O} + \eta_{\mathrm{sign}}$$

be the public-random exceptional-key probabilities for the fast projective and graph-direction equivalent-key theorems. Total variation is applied only to these key-level events, not to a bounded attack execution.

Corollary G.5 (Official seeded recovery in ideal-primitive models)**.** Under the official seeded key generation:

1. with SHAKE modeled as an ideal XOF, the probability that a key fails the graph-direction equivalent-key hypotheses is at most $\eta_{\mathrm{local}} + \delta_{\mathrm{XOF}}$, while the corresponding fast-projective bound is $\eta_{\mathrm{fast}} + \delta_{\mathrm{XOF}}$;
2. with AES modeled as an ideal cipher, the corresponding bounds are $\eta_{\mathrm{local}} + \delta_{\mathrm{IC}}$ and $\eta_{\mathrm{fast}} + \delta_{\mathrm{IC}}$.

Every other key for the relevant route satisfies the corresponding key-by-key recovery theorem: accepted outputs are always correct, and verified restart terminates almost surely with the expected-work bounds of Theorems B.16 and B.18. At Levels I, III, and V, both ideal-key exceptions are negligible compared with the transfer terms, so the base-two logarithms round respectively to

$$(-121.245, -184.715, -248.286) \quad \text{and} \quad (-99.863, -96.655, -94.112)$$

for SHAKE and AES.

Proof. By Propositions G.1 and G.2, the probability of every key event changes by at most the corresponding total-variation distance. Apply this separately to the graph-direction event of Corollary B.17 and the fast-projective event of Corollary B.19. On either complement, the corresponding theorem is key by key and independent of how the expanded key was sampled. Exact verification is distribution-free. The ideal exceptional probabilities are far below the transfer terms at the displayed precision. $\square$

G.5 Why this does not prove a fixed-primitive theorem

Proposition G.6 (Support barrier for deterministic seeded expansion)**.** Let $E : \{0,1\}^\lambda \to \Omega$ be any deterministic public expansion map, let S be uniform on $\{0,1\}^\lambda$, and let U be uniform on the finite set Ω. Then

$$\Delta_{\mathrm{TV}}(E(S), U) \geq 1 - \frac{2^\lambda}{|\Omega|}.$$

In particular, if $\Omega = \{0,1\}^B$, the distance is at least $1 - 2^{\lambda - B}$.

Proof. The distribution $E(S)$ is supported on at most 2^λ points. Let R be its support. Then $\Pr[E(S) \in R] = 1$, while $\Pr[U \in R] = |R|/|\Omega| \leq 2^\lambda/|\Omega|$. The variational characterization of total variation gives the claim. $\square$

Thus the fixed public SHAKE/AES stream vector cannot be information-theoretically close to independent uniform streams. Nor is there a generic public-seed PRG reduction: the public seed is revealed, expansion is deterministic, and a distinguisher can recompute the official stream and compare it with the transcript. The ideal-XOF and ideal-cipher key-density conclusions above are exact statements inside those models; they do not instantiate themselves for fixed primitives.

The key-by-key recovery theorem nevertheless applies to every fixed-primitive key satisfying the key-by-key conditions. A high-probability standard-model statement must therefore analyze the relevant algebraic bad events under the actual fixed expansion map, or introduce an explicitly event-specific hardness assumption. The support barrier does not rule out such work; it rules out a generic statistical or transcript-pseudorandomness transfer that replaces the revealed deterministic expansion by independent uniform streams.

H Optional identification with the planted oil space

Neither recovery theorem requires uniqueness: it keeps every verified candidate and searches all of them for a signing witness. This appendix proves the stronger generic identification statement that, for almost every key, no second K-rational common isotropic O-space exists. Fix the planted oil space $\mathcal{O}_0$. Let $\mathcal{O}' \neq \mathcal{O}_0$ be another O-space and put

$$s = O - \dim(\mathcal{O}' \cap \mathcal{O}_0), \qquad 1 \leq s \leq O.$$

Lemma H.1 (Relative Grassmannian stratum)**.** The locus of such $\mathcal{O}'$ with fixed s has dimension $s(V+O-s)$.

Proof. Choose the $(O-s)$-dimensional intersection inside $\mathcal{O}_0$, the s-dimensional image of $\mathcal{O}'$ in $K^{V+O}/\mathcal{O}_0 \simeq K^V$, and the graph map back to the s-dimensional quotient of $\mathcal{O}_0$. The dimensions are $(O-s)s$, $s(V-s)$, and s^2, respectively. ☐

Lemma H.2 (Independent extra isotropy conditions)**.** For fixed $\mathcal{O}'$ in this stratum, requiring one central form that already vanishes on $\mathcal{O}_0 \times \mathcal{O}_0$ also to vanish on $\mathcal{O}' \times \mathcal{O}'$ imposes exactly

$$c_s = \binom{O+1}{2} - \binom{O-s+1}{2} = \frac{s(2O-s+1)}{2}$$

additional independent linear conditions.

Proof. Restriction to $\mathcal{O}'$ already vanishes on $\mathrm{Sym}^2(\mathcal{O}' \cap \mathcal{O}_0)$. Conversely, after choosing complements, every symmetric form on $\mathcal{O}'$ that vanishes on this subspace extends to an ambient symmetric form vanishing on $\mathcal{O}_0 \times \mathcal{O}_0$. The restriction map therefore has the displayed rank. ☐

Theorem H.3 (Uniqueness codimension and rational first moment)**.** For m independent central forms, the key locus admitting an alternative common isotropic O-space of type s has codimension at least

$$\delta_s = m\frac{s(2O-s+1)}{2} - s(V+O-s).$$

The minimum over s equals $903, 1927, 3539$ for Levels I, III, and V. With Gaussian binomial coefficients denoted by $\begin{bmatrix} a \\ b \end{bmatrix}_Q$,

$$\Pr[\text{a second } K\text{-rational oil space}] \leq \sum_{s=1}^{O} \begin{bmatrix} O \\ s \end{bmatrix}_Q \begin{bmatrix} V \\ s \end{bmatrix}_Q Q^{s^2-mc_s}.$$

Writing the displayed first-moment bound as η_{uniq}, we have

$$\log_2 \eta_{\mathrm{uniq}} < -18932.346, \quad -40401.586, \quad -74198.865$$

at Levels I, III, and V, respectively.

Proof. Over the stratum of dimension $s(V+O-s)$, each of the m form fibers loses c_s dimensions by Lemma H.2. The incidence variety therefore has codimension at least $mc_s - s(V+O-s)$ after projection to key space. The function δ_s is concave in s, so its minimum on $1 \leq s \leq O$ occurs at an endpoint; direct evaluation gives the displayed values and shows that $s=1$ minimizes in all three parameter sets.

For the finite-field statement, the number of K-rational O-spaces in the type-s stratum is

$$\begin{bmatrix} O \\ s \end{bmatrix}_Q \begin{bmatrix} V \\ s \end{bmatrix}_Q Q^{s^2}.$$

For each fixed $\mathcal{O}'$, the m independent forms satisfy the c_s extra conditions with probability Q^{-mc_s}. The union bound gives the formula; exact evaluation gives the stated logarithms. ☐

Corollary H.4 (Generic identification with the planted decomposition)**.** In the public-random model, outside the exceptional event of Corollary B.17 and the additional event of probability η_{uniq} in Theorem H.3, every exact graph-direction candidate has the planted oil space. The same conclusion holds for the fast projective route outside the exceptional event of Corollary B.19 and the uniqueness event. In either case, the returned decomposition is the planted expanded decomposition up to the verified coordinate representation. The secret seed is still not recovered or needed.

Proof. Every exact candidate defines a common isotropic O-space. Uniqueness forces that space to be $\mathcal{O}_0$, and its graph determines the expanded decomposition used by Proposition 3.4. $\square$

I Alternative elimination and low-space directions

The main text uses complete univariate resolutions because they are the only route for which we have both a suitable theorem and full failure accounting. This appendix records two exact ways to extract a root more compactly once a quotient algebra exists, and then explains why several natural attempts do not yet improve the proved exponent.

I.1 Extracting an isolated root from an existing quotient algebra

Let $A = L[y_1, \ldots, y_V]/I$ be the reduced affine coordinate algebra of a square core, of degree $d \leq D$. Write Z for its geometric root set, and let $q_1, \ldots, q_m$ denote the original restricted equations in A.

Proposition I.1 (Norm-gradient root formula)**.** Suppose $p \in L^V$ is the unique point of Z satisfying all m original equations. Choose $\rho \leftarrow L^m$, put $h = \sum_i \rho_i q_i \in A$, and define

$$\mathcal{N}_h(z_0, \ldots, z_V) := \mathrm{Norm}_{A/L}\left(h + z_0 + \sum_{j=1}^{V} z_j y_j\right). \tag{69}$$

With probability at least $1 - (d-1)/|L|$, h vanishes at p and at no other geometric core point. On that event,

$$\partial_{z_0}\mathcal{N}_h(0) \neq 0, \qquad p_j = \frac{\partial_{z_j}\mathcal{N}_h(0)}{\partial_{z_0}\mathcal{N}_h(0)} \quad (1 \leq j \leq V). \tag{70}$$

Equivalently, if M_h and M_{y_j} are the multiplication operators in A, then $\ker M_h$ is one-dimensional and every nonzero $w \in \ker M_h$ satisfies $M_{y_j} w = p_j w$.

Proof. After scalar extension to an algebraic closure, reducedness gives

$$A \otimes_L \overline{L} \cong \prod_{x \in Z} \overline{L}.$$

At every $x \neq p$, the vector $(q_1(x), \ldots, q_m(x))$ is nonzero. The condition $h(x) = 0$ cuts out a proper L-linear subspace of L^m, so a uniform ρ satisfies it with probability at most $1/|L|$. A union bound over at most $d-1$ points proves the selector bound. On the good event, the norm factors as

$$\mathcal{N}_h(z) = \prod_{x \in Z}\left(h(x) + z_0 + \sum_{j=1}^{V} z_j x_j\right).$$

Differentiating at the origin gives

$$\partial_{z_0}\mathcal{N}_h(0) = \prod_{x \neq p} h(x), \qquad \partial_{z_j}\mathcal{N}_h(0) = p_j \prod_{x \neq p} h(x),$$

which proves (70). In the same product decomposition, M_h is diagonal with a unique zero eigenvalue and the coordinate operators act on its kernel by the scalars p_j. $\square$

Remark I.2 (Several surviving roots)*.* For an arbitrary selector h, let $S = \{x \in Z : h(x) = 0\}$. The lowest nonzero homogeneous part of (69) at the origin is

$$\left(\prod_{x \notin S} h(x)\right) \prod_{x \in S}\left(z_0 + \sum_{j=1}^{V} z_j x_j\right).$$

Thus the first nonzero norm jet is the Chow product of all selected roots. The linear formula (70) is exactly the case $|S| = 1$.

Proposition I.3 (Resultant-gradient root formula)**.** Assume $\operatorname{char} L \neq 2$. Let $F_0, \dots, F_V \in L[t_0, \dots, t_V]$ be homogeneous quadrics with a unique geometric common point $p = [p_0 : \cdots : p_V]$. Assume that their value Jacobian on the tangent space $T_p\mathbb{P}^V$ has rank V. Let $\mathcal{R}$ be their multivariate resultant, written in the monomial coefficients $c_{i,\alpha}$ of F_i, $|\alpha| = 2$. Then the coefficient tuple is a smooth point of the resultant hypersurface. If $\lambda = (\lambda_0, \dots, \lambda_V)$ spans the left kernel of the tangent Jacobian, there is a scalar $\kappa \neq 0$ such that

$$\frac{\partial \mathcal{R}}{\partial c_{i,\alpha}} = \kappa \lambda_i p^\alpha \qquad (0 \leq i \leq V,\ |\alpha| = 2). \tag{71}$$

In particular, for any i, k with $\lambda_i p_k \neq 0$,

$$\frac{p_j}{p_k} = \frac{\partial \mathcal{R}/\partial c_{i,e_j+e_k}}{\partial \mathcal{R}/\partial c_{i,2e_k}} \qquad (0 \leq j \leq V). \tag{72}$$

The coefficient convention in (72) treats $t_j t_k$ as one monomial, with no symmetric-matrix factor of two.

Proof. Consider the incidence variety of coefficient tuples and projective points satisfying $F_i(p) = 0$ for every i. Its first-order equations at the given pair are

$$\delta F_i(p) + dF_i(p)[\delta p] = 0 \qquad (0 \leq i \leq V).$$

The tangent Jacobian has rank V, so a tangent displacement δp exists exactly when

$$\sum_{i=0}^{V} \lambda_i \, \delta F_i(p) = 0.$$

Uniqueness and transversality make the incidence projection locally one-to-one onto a smooth branch of the resultant hypersurface. Its tangent hyperplane is therefore the kernel of the displayed nonzero functional. The differential of the irreducible resultant is proportional to that functional, giving (71); the ratio formula follows by evaluating the two quadratic monomials at p. □

Remark I.4 (Compression achieved, evaluator still missing)*.* The norm gradient outputs one root using only $V + 1$ field elements, and the resultant gradient uses one nonzero coefficient block. If quotient multiplication is available as a black box, a one-dimensional kernel can be found with $O(d)$ stored field elements and $O(d)$ multiplication-operator products by standard black-box linear algebra [23]. Alternatively, Baur–Strassen converts any arithmetic circuit for the norm or resultant into a circuit for all first derivatives with constant-factor overhead [16]. These statements compress the *root-extraction interface*; they do not construct the degree-d quotient algebra, multiplication operators, norm, or resultant in $o(d)$ space. With current constructions, the time remains near-quadratic in d (with the fixed ε loss of Proposition B.9) and at least one degree-d state vector remains. Moreover, Proposition A.3 gives no production-parameter density bound for uniqueness of the full-system root. The formulas are therefore exact algorithmic targets, not a practical attack.

I.2 Why the spare equations do not yet lower the finite exponent

The full restricted system has $e = m - V \in \{2, 2, 3\}$ equations beyond the square core, and generically those equations isolate the planted point. We tested whether they also lower the solving exponent. The following deliberately optimistic semi-regular screen uses the Hilbert series

$$\frac{(1 - t^2)^{V+e}}{(1 - t)^{V+1}}$$

for $V + e$ homogeneous quadrics in $V + 1$ projective variables. Because the actual system contains a planted point, this is a diagnostic rather than a theorem. At the last positive coefficient, the monomial counts are:

Level	V	degree	$\log_2 N$	optimistic $\log_2 N^2$	current $\log_2 D^2$
I	52	27	69.787	139.574	104
III	76	39	102.585	205.170	152
V	102	48	131.814	263.629	204

Even charging only N^2 sparse black-box work is worse than the baseline D^2 geometric scale and also worse than the primary $\varepsilon = 0.3$ fast-line indicators after field and circuit conversion; dense linear algebra is worse still. This does not rule out a coefficient-specific method, but it rules out the most direct overdetermined-Macaulay improvement under standard linear algebra.

Proposition I.5 (Underslicing loses more in hit probability than it saves in degree)**.** Let $0 \le d < V$, and choose a uniform projective d-plane over $K = \mathbb{F}_Q$. Its probability of meeting the planted projective oil space is

$$\pi_d = 1 - \prod_{i=0}^{d} \frac{1 - Q^{i-V}}{1 - Q^{i-(V+O)}}. \tag{73}$$

If the same degree-squared square-core solver is run independently on every sampled plane, its expected leading degree factor is $4^d/\pi_d$. For $d = V - r$ and the QR-UOV base field $Q = 127^3$, removing $r = 1, 2, 3$ slice dimensions increases the base-two exponent relative to the $d = V$ slice by

$$18.966053, \qquad 37.932107, \qquad 56.898161$$

bits, respectively, before any other work is charged. Choosing the slices over the larger solver field only worsens this tradeoff.

Proof. Write $k = d + 1$. A vector k-subspace disjoint from the fixed O-dimensional oil space is the graph of a linear map from a k-subspace of the V-dimensional quotient to the oil space. Hence there are

$$Q^{Ok} \begin{bmatrix} V \\ k \end{bmatrix}_Q$$

disjoint subspaces, out of $\begin{bmatrix} V+O \\ k \end{bmatrix}_Q$ total. Taking the ratio and expanding the Gaussian binomial coefficients gives (73). Independent repetition needs expected $1/\pi_d$ planes. Substitution into $4^d/\pi_d$ gives the displayed values, which are recomputed by the companion numerical audit. Asymptotically each removed dimension costs $\log_2 Q - 2 = 18.966054\ldots$ bits: it saves two degree-squared bits but loses roughly one factor of Q in intersection probability. $\square$

The same calculation covers the equivalent hybrid strategy of guessing one base-field coordinate of the oil point before solving: every guessed coordinate costs roughly $\log_2 Q$ bits and saves only two bits in the degree-squared solver. Thus the minimal guaranteed slice is also the best random underslice under the current solver model.

The other natural routes also stop short of an exponent improvement. Explicit resultant matrices remain exponentially larger than the D-dimensional quotient representation in the dense quadratic regime [17]. Chow-form algorithms are polynomial in geometric degree but do not provide a subquadratic dependence specialized to these parameters [18]. Faster modular composition lowers an operation used after a univariate algebra exists, not the geometric-intersection stage that constructs that algebra [19].

We also checked two recent refinements of Macaulay/Gröbner solving. Degree smoothing replaces a full Macaulay matrix by a sufficient submatrix lying between two adjacent integer degrees [20]. Our optimistic screen already charges the complete monomial space at the *lower* adjacent degree and is still 35.574, 53.170, and 59.629 bits above the D^2 geometric scale; smoothing within that interval cannot reverse the comparison. The recent F4-tracer analysis for generic square degree-δ systems has leading arithmetic scale $\widetilde{O}(\delta^{\omega V+1}e^{cV})$ and relies on trace compatibility and genericity conjectures [21]. For quadrics, even its factor $2^{\omega V}$ exceeds $D^2 = 2^{2V}$ for every currently proved $\omega > 2$. Thus these improvements are valuable in their intended regimes but do not lower the exponent here. The norm and resultant gradients identify a smaller *output* target; no proved route found in this audit computes that target below the published near-quadratic-in-D line used here.

J Reduced-parameter end-to-end instantiation

The artifact now contains an executable instantiation of the complete attack rather than only independent checks of its ingredients. The reduced scheme follows the QR-UOV algebraic construction in the official specification [2]: central symmetric matrices over an extension field have a zero oil–oil block, the secret change of variables has the compact form

$$S = \begin{pmatrix} I & G \\ 0 & I \end{pmatrix},$$

and the extension-field matrices are pulled back to base-field quadratic forms through a nonzero linear functional, implemented here by the trace pairing. Seed expansion, byte serialization, and the exact domain-separated hash interface are deliberately simplified; the public quadratic relation attacked by the algorithm is unchanged.

The default demonstration uses

$$q = 13, \qquad \ell = 3, \qquad V = 3, \qquad M = 2,$$

so the expanded UOV dimensions are

$$v = 9, \qquad m = o = 6, \qquad n = 15.$$

The residual projective core has $a = m - 4 = 2$, hence $C = 4$, $C_0 = 8$, and precision $P = 4C_0 + 1 = 33$. The attack function receives only the expanded public matrices, message, and salt. It executes target normalization, the anchored Chevalley–Warning construction, a common totally isotropic subspace for three forms, affine charting, all four coefficientwise homotopy lifts, local-separator recombination, rational reconstruction, valuation-safe specialization, exact Frobenius descent, one coordinate rerun, affine rescaling, and unchanged public verification.

For the canonical seed 20260731, the program first generated an honestly verifiable signature and then produced the forged signature

$$(0, 8, 2, 10, 0, 0, 0, 4, 10, 5, 11, 8, 0, 12, 1) \in \mathbb{F}_{13}^{15}$$

for the recorded message and salt. The public verifier accepted it. The successful run used the trace in Table 5.

Table 5: Canonical reduced-parameter attack trace. Wall time is machine-dependent and is not used in the security ledger.

Quantity	Value
Salt attempts / attack records	1/1
Target eliminant degree	4
Branch lifts	8 (4 initial, 4 coordinate rerun)
Checked lifted-branch residuals	4
Rational reconstructions	31
Coordinate reruns	1
Target roots examined / simple	2/1
Frobenius passes / exact verified candidates	1/1
Exhaustive $\mathbb{F}_{13}$ / $\mathbb{F}_{13^2}$ residual roots	2/2
Python wall time	0.415 seconds

The exhaustive residual-root enumeration is deliberately performed only *after* the local-separator algorithm has selected and reconstructed its candidate. It is not a solver subroutine. It checks that every extension-field root appears in the specialized eliminant, that every simple eliminant root has a singleton fiber, and that the accepted point is among the base-field roots.

The canonical transcript records the public key, target, isotropic subspace, chart, affine core, lifted-system parameters, separator, specialized characteristic polynomial and scalar polars, accepted root, recovered coordinates, signature, and operation counts. A separate public replay program consumes that transcript without the secret key, reconstructs the affine core, reruns the complete local-separator solver with the recorded separator, and reproduces the accepted forgery.

A fixed regression suite over the five consecutive seeds $20260000, \ldots, 20260004$ produced five publicly valid forgeries and five successful public replays. At the imposed finite retry limits, the largest salt count was two and the slowest individual run took 10.252 seconds. This is a regression test, not a statistical estimate of the asymptotic success probability. A supplementary run over $q = 29$ with the same dimensions also forged and replayed successfully.

This closes the narrow composability objection: the artifact now instantiates one finite end-to-end attack on reduced QR-UOV instances. It does *not* establish the Level-I gate count. The prototype uses naive truncated-series arithmetic at $a = 2$ and therefore neither implements nor benchmarks the fast dyadic multiplication schedule assumed by the full-parameter ledger. Full parameters remain infeasible to execute, and the public-random-model, seeded-transfer, characteristic-polynomial degree, and memory-model caveats remain.

K Archived validation, assessment, and audit inventory

This section preserves the detailed validation and audit record consolidated from the main paper.

K.1 Validation and remaining risk

The bundle includes:

1. `QRUOV_local_separator_audit.py`, which computes the optimized degree, finite probabilities, and all component costs;
2. `verify_local_separator_ledger.py`, a separately written implementation importing neither the primary audit nor its parent modules, matching all displayed totals to below $5 \cdot 10^{-10}$ bits;

3. the end-to-end reduced attack `implementation/qruov_local_separator_forgery.py`, its public replay checker, five-instance regression suite, and complete public transcripts;
4. Rabin certificates for every newly used sparse modulus;
5. the original split-before-lift toys validating branch lifting, symmetric recombination, and Frobenius arithmetic, together with an exhaustive $\mathbb{F}_9/\mathbb{F}_3$ toy showing that a non-base point passes the one-sketch test exactly on separator collisions; and
6. `toy_infinite_branch_specialization.py`, which exhibits the subtle case in which an escaping-branch numerator term survives normalization but is a multiple of the finite eliminant and hence vanishes at every finite simple root.

The principal remaining proof obligations are external checks that the characteristic-polynomial degree theorem in Lemmas 8.2 and 8.3 applies to the exact saturated dominant homotopy algebra and to the rational-filter graph with the stated degree bounds. The determinant identities are exact and no defect is currently known, but this identification with the parent solver's saturated dominant algebra is what the degree-8 retuning depends on. The reduced implementation supplies a literal finite schedule for the toy instance, but the dyadic fast online-product envelope used at Level I still needs a line-by-line full-scale algorithmic derivation; the multiplier table shows exactly how much error that part can absorb. The earlier caveats about the idealized public-random model, seeded transfer, and astronomical memory traffic are unchanged.

K.2 Assessment

This is a real strengthening rather than a cosmetic ledger improvement. The preceding candidate crossed Level I by splitting before lifting; the present argument removes the largest remaining auxiliary-field overkill. It replaces a global primitive-element event with a one-sided local isolation event and turns Frobenius filtering from probabilistic to exact. Under the inherited model, the cumulative improvement over the strict parent bundle is 11.254751 bits.

The result is now robust to far larger rational-reconstruction constants than the original 1.87-bit-margin version. The reduced implementation materially strengthens the evidence: the claimed transformations and filters have now been composed into an actual public-key forgery rather than tested only in isolation. This manuscript distinguishes this demonstrated toy-scale attack from the Level-I extrapolation and should await specialist verification that the characteristic-polynomial degree theorem applies to the exact saturated dominant algebra and its rational-filter graph.

K.3 Executable audits

The bundle contains the following executable checks and an end-to-end reduced-parameter attack.

Combined numerical audit. `QRUOV_scissors_audit.py` recomputes the Chevalley–Warning slacks, the projective $m-4$ cores, fixed-key seed–oil intersection bound, second-moment nonsingular-root bound, chart and separator probabilities, Karatsuba field circuits, direct and directional cost rows, memory envelopes, and attack-specific retuning thresholds. It also applies Rabin's criterion to every sparse auxiliary-field polynomial used in the official and threshold rows. The script emits both human-readable and machine-readable output.

Directional algebra audit. `QRUOV_directional_audit.py` checks the corrected $2C_0$ precision, Vandermonde product starts, characteristic-127 lifting identities, flattened bivariate multiplication, sparse-field certificates, the recursive Karatsuba plans, and the complete graph-recovery pipeline on 200 random toy instances. Of those instances, 199 had a full-rank planted direction and all 199 yielded the exact graph; the remaining direction was correctly rejected for retry.

Exact degree-three direct-forgery toy. `QRUOV_direct_qr_toy.py` generates independent extension-field UOV forms over $\mathbb{F}_{3^3}$, hides their oil space, pulls each form back by a trace to a scalar base-field quadric, and performs full target normalization, random seed-subspace search, Chevalley–Warning extension, projective residual search, affine rescaling, and public verification. In the frozen 200-key run, 199 keys yielded a verified forgery within twelve trials. Every returned vector passed two independent evaluators; 185 final seed subspaces were disjoint from the hidden oil space, and every selected seed point was off oil. The experiment checks the attack logic; it is not a full-parameter benchmark.

End-to-end local-separator forgery. `implementation/qruov_local_separator_forgery.py` instantiates the complete reduced attack described in Section J. Its public replay checker reconstructs the residual system and reproduces the forgery without secret material. The frozen regression suite succeeds on five of five instances, and the supplementary $q = 29$ instance also succeeds. Exhaustive root enumeration is used only after acceptance as a validation oracle, never as the attack solver.

L Reproducibility and executable attack description

L.1 Companion executable audits

The bundle contains four standard-library Python programs.

- `QRUOV_scissors_audit.py` recomputes the target-orthogonal dimension inequalities, projective $m-4$ cores, fixed-key seed–oil intersection bounds, second moments, eligible-root probabilities, random-chart and separator factors, both finite-size cost rows, memory envelopes, Karatsuba plans, sparse-field certificates, and attack-specific retuning diagnostics. It emits text and JSON and freezes every displayed number.
- `QRUOV_directional_audit.py` checks the corrected $2C_0$ reconstruction precision, Vandermonde product starts, characteristic-127 lifting identities, flattened multiplication, sparse-field certificates, recursive Karatsuba convolution, and complete graph recovery on 200 toy keys.
- `QRUOV_direct_qr_toy.py` generates exact degree-three extension-field UOV instances, applies the trace pull-back, and executes full target normalization, seed search, Chevalley–Warning extension, projective residual search, affine rescaling, and public verification end to end.
- `QRUOV_cw_helper.py` supplies the finite-field and Chevalley–Warning helper routines used by the direct experiment.

These programs audit formulas and attack logic. They do not instantiate the astronomically large full-parameter symbolic state.

L.2 Executable attack description

Projective target-orthogonal direct-forgery route.

1. Derive the official nonzero verification target y from the message and salt. Choose a public invertible output transformation sending y to $(0, \dots, 0, 1)$, and apply the same transformation to the public quadrics.

2. Sample a uniform seven-dimensional seed subspace $S_0 \subset \mathbb{F}_q^n$. Enumerate $\mathbb{P}(S_0)$ until finding a nonzero common zero of the first three transformed quadrics; Chevalley–Warning guarantees one.
3. Starting from its span, repeatedly intersect the three polar-orthogonal spaces and solve the induced three quadrics on a seven-dimensional quotient slice. Extend until obtaining a common totally isotropic subspace U of dimension $m - 3$.
4. Choose a random nonzero chart form on U. Restrict the next $m - 4$ zero-target quadrics to the chart and run the checked finite-characteristic nonsingular-root solver on the resulting $m - 4$ variable square system.
5. At every represented rational projective root, evaluate the final transformed quadratic. If its value is a nonzero square, recover either affine scale making that value one, reverse the transformations, and re-evaluate all original public equations.
6. Return the first vector passing the unchanged QR-UOV verifier. If the chart, solver invocation, or residual search fails, change the salt and attack coins and repeat. Exact verification gives one-sided error.

Primary finite-characteristic direction route.

1. Expand the public key to matrices P_i over $K = \mathbb{F}_{127^3}$. Choose a nonzero public direction $c \in K^O$ and form $q_{i,c}(x) = P_i(-x, c)$.
2. Choose a random $V \times m$ output-recombination matrix R and form the square system $f = Rq_c$. Work in $K' = \mathbb{F}_{127^d}$ using the sparse degree-18, 24, or 30 model and the audited recursive Karatsuba circuit for the relevant parameter level.
3. Run the finite-characteristic nonsingular-root algorithm of Theorem 7.1, using the Vandermonde product start and a full random separator. A bad invocation may omit roots or fail; it cannot create an accepted false key.
4. Re-evaluate all m original directional equations at every represented K-rational root. At each survivor, solve the O public linear systems of Proposition 5.2. Keep only graphs passing every graph identity, total-isotropy check, inverse pull-back, and exact public-map factorization.
5. Run the basis-first signing scan and, if needed, the pair-line fallback. Return the first exact candidate with an invertible oil matrix. If no candidate passes, sample fresh solver coins and then a fresh outer direction.

Coordinate directions and Wronskian condensers provide a deterministic outer fallback; the primary finite-size ledger uses the simpler random full-rank recombination analyzed in Theorem 7.3.

Fast projective route.

1. Set $\varepsilon = 0.3$. Use the minimum fields $s = 8, 12, 15$ for expected-restart analysis or the safety fields $s = 9, 13, 16$ for the one-invocation theorem. Sample a full-rank slice matrix M and an output recombination R over L_s; restrict the public quadrics and form the square core.
2. Run one finite checked invocation of the van der Hoeven–Lecerf solver. Return `fail` on an exceptional division, degree mismatch, or consistency failure. Do not hide an unbounded internal restart inside one outer draw.
3. Canonicalize and verify the resolution. Evaluate all original restricted equations in its quotient algebra, prefilter with Proposition B.14, extract L_s-rational roots, and recheck all equations at each reconstructed point.

4. At every survivor, compute the common derivative kernel. Require dimension O, descent to a graph over K, total isotropy, all graph identities, the official inverse pull-back, and exact factorization of the original public map.
5. Run the basis-first signing scan and, if needed, the pair-line fallback. Return the first verified signing certificate; otherwise begin a fresh outer trial.

Both wrappers have one-sided error: a solver failure may cause a retry, but exact public verification prevents an invalid equivalent key or invalid signature from being returned. The output is a verified expanded signing decomposition and an inverse oil matrix, not the original seed or an official serialized private key.

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